MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

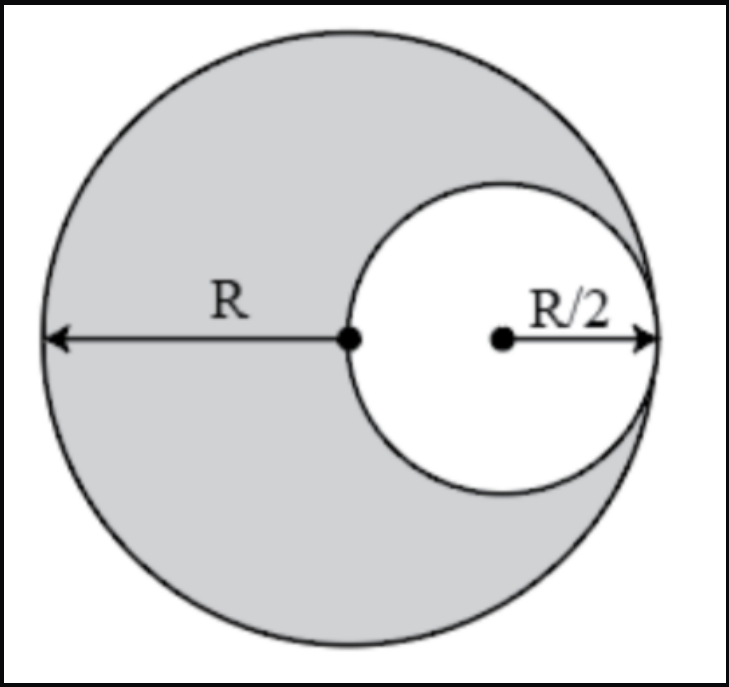
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A spherical hole is made in a solid sphere of radius R. The mass of the sphere before hollowing was M. The gravitational field at the centre of the hole due to the remaining mass is



(A) Zero

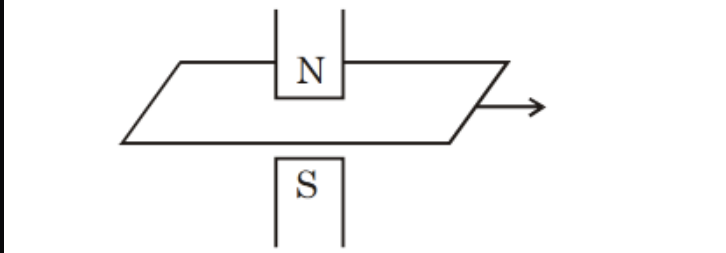
(B) $\frac{GM}{8R^2}$

(C) $\frac{GM}{2R^2}$

(D) $\frac{GM}{R^2}$

Q.2

A metal sheet is being pulled out of a region of magnetic field. If the sheet is non magnetic :-



(A) No force is required and no heat is dissipated in sheet.

(B) Force is required but no heat is dissipated in sheet.

(C) No force is required but heat is dissipated in the sheet.

(D) Force is required and heat is dissipated in the sheet.

Q.3

At room temperature, the rms speed of the molecules of a certain diatomic gas is found to be $1930\ \mathrm{ms}^{-1}$. The gas is

(A) H_2

(B) F_2

(C) O_2

(D) Cl_2

Q.4

In a nuclear fission, 0.1% mass is converted into energy. The energy released by the fission of $1\ \mathrm{kg}$ mass is

(A) $9 \times 10^{13}\ \mathrm{J}$

(B) $9 \times 9^{19}\ \mathrm{J}$

(C) $9 \times 10^{16}\ \mathrm{J}$

(D) $9 \times 10^{17}\ \mathrm{J}$

Q.5

If n drop, each of capacitance C, coalesce to form a single big drop, then the ratio of the energy stored in the big drop to that in each small drop will be

(A) $n : 1$

(B) $n^{1/3} : 1$

(C) $n^{5/3} : 1$

(D) $n^2 : 1$

Q.6

Assertion- The red light is just able to emit photoelectrons from a metal surface, then violet light cannot emit the photoelectrons from that metal.
Reason- The energy of violet light photons is more than red light photons.

(A) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.

(B) Assertion is true but Reason is false.

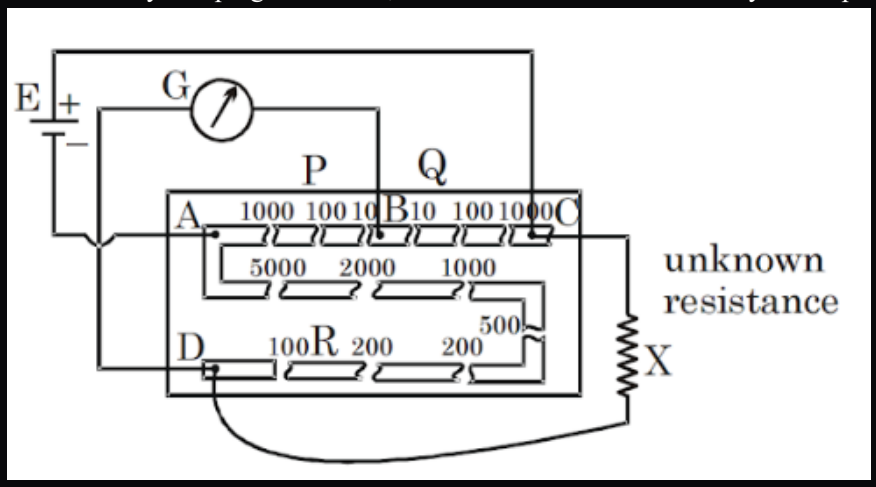
(C) Assertion is false but Reason is true.

(D) Both Assertion and Reason are false.

Q.7

In the figure shown below, the maximum and minimum possible unknown resistance (X), that can be measured by the post office box are $X_{\max}$ and $X_{\min}$ respectively. Then the ratio of these two resistances is : (In this experiment, we take out only one plug in arm

AB and only one plug in arm BC, but in arm AD we can take any no. of plugs):



(A) 3×10^5

(B) 5×10^5

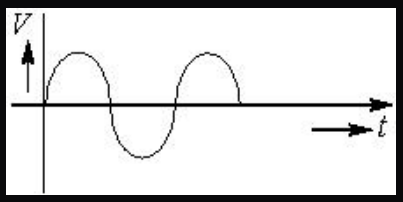
(C) 7×10^5

(D) 9×10^5

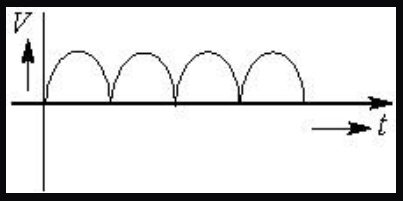
Q.8

Which of the following speed-time graphs exist in the nature?

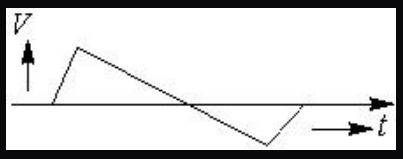
(A)



(B)



(C)



(D) All of the above

Q.9

Two long parallel wires P and Q are held perpendicular to the plane of the paper at a separation of $5\ \mathrm{m}$. If P and Q carry currents of $2.5\ \mathrm{A}$ and $5\ \mathrm{A}$ respectively in the same direction, then the magnetic field at a point midway between P and Q is

(A) $\frac{\mu_0}{\pi}$

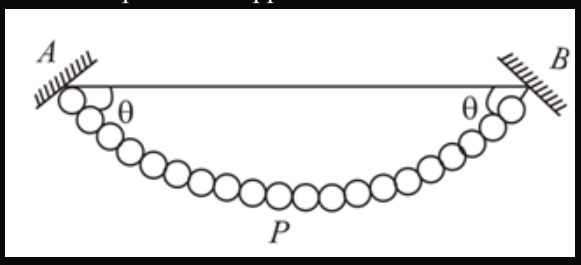
(B) $\sqrt{3}\frac{\mu_0}{\pi}$

(C) $\frac{\mu_0}{2\pi}$

(D) $\frac{3\mu_0}{2\pi}$

Q.10

A flexible chain of mass M hangs between two fixed points A and B at the same level. The inclination of the chain with the horizontal at the two points of support is θ. The tension at the mid-point of the chain is :



(A) $\frac{Mg\cot\theta}{2}$

(B) $Mg\cot\theta$

(C) Zero

(D) $\frac{Mg}{2}\left(\sin\theta+\cos\theta\right)$

Q.11

Two beams of light have intensities of I_0 and $4I_0$ in double slit experiment. The phase difference between the beams is $\frac{\pi}{3}$ at point A and $\frac{\pi}{2}$ at point B. Then, the difference between the resultant intensities at A and B is

(A) $2I_0$

(B) I_0

(C) $4I_0$

(D) $5I_0$

Q.12

A brass rod of length 500 mm and diameter 3 mm is joined to a steel rod of same length and diameter at 50 $^{\circ}C$. If the coefficients of linear expansion of brass and steel are $2.5\times10^{-5}\ ^{\circ}C^{-1}$ and $1.25\times10^{-5}\ ^{\circ}C^{-1}$, then change in length of the combined rod at 200$^{\circ}C$ is

(A) 2.4 mm

(B) 2.8 mm

(C) 3.2 mm

(D) 3.6 mm

Q.13

The specific heats of an ideal gas at constant pressure and constant volume are $525\ \mathrm{J\ kg^{\circ}C^{-1}}$ and $315\ \mathrm{J\ kg^{\circ}C^{-1}}$ respectively. Its density at NTP is

(A) $0.64\ \mathrm{kg\ m^{-3}}$

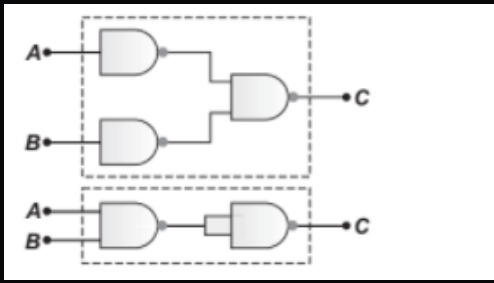
(B) $1.20\ \mathrm{kg\ m^{-3}}$

(C) $1.75\ \mathrm{kg\ m^{-3}}$

(D) $2.62\ \mathrm{kg\ m^{-3}}$

Q.14

The combination of NAND gates shown here under figure, are equivalent to



(A) an OR gate and an AND gate respectively

(B) an AND gate and a NOT gate respectively

(C) an AND gate and an OR gate respectively

(D) an OR gate and a NOT gate respectively

Q.15

A current of $4\ \mathrm{A}$ flows in a coil when connected to a $12\ \mathrm{Vdc}$ source. If the same coil is connected to a $12\ \mathrm{V}, 50\ \mathrm{rad/s}$ a.c. source, a current of $2.4\ \mathrm{A}$ flows in the circuit. Determine the inductance of the coil.

(A) $0.08\mathrm{H}$

(B) $0.04\mathrm{H}$

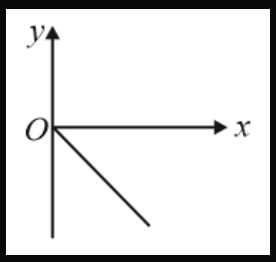
(C) $0.02\mathrm{H}$

(D) $1\mathrm{H}$

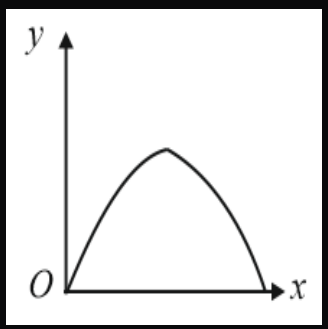
Q.16

Two particles of mass m and $2m$ have their position vectors as a function of time as $r_1(t) = t\,\hat{\mathrm{i}} - t^3\,\hat{\mathrm{j}} + 2t^2\,\hat{\mathrm{k}}$ and $r_2(t) = t\,\hat{\mathrm{i}} - t^3\,\hat{\mathrm{j}} - t^2\,\hat{\mathrm{k}}$ respectively (where t is the time). Which one of the following graphs represents the path of the centre of mass?

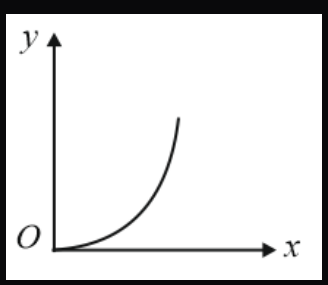
(A)



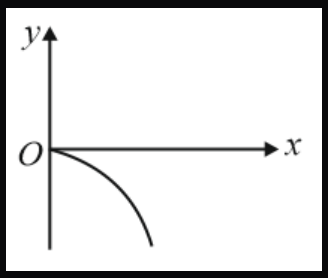
(B)



(C)

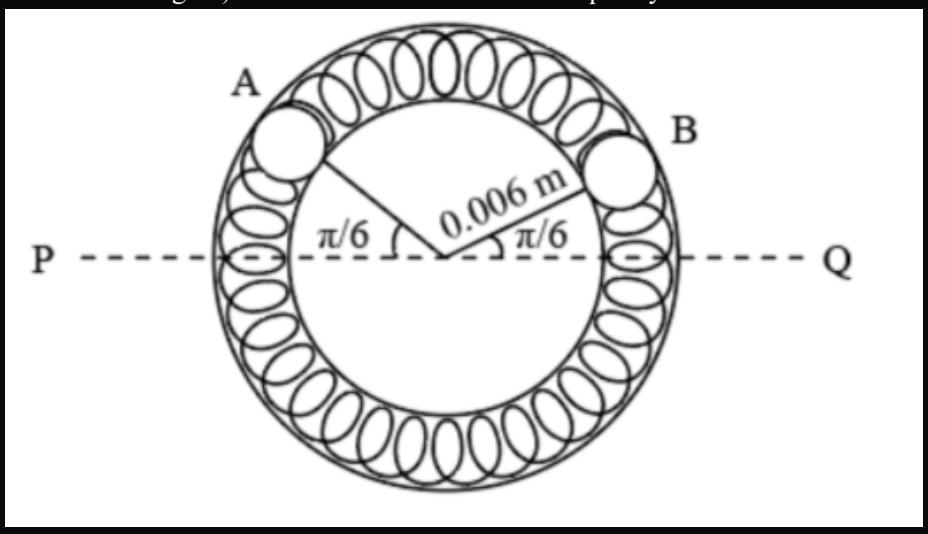


(D)



Q.17

Two identical balls A and B, each of mass $0.1\ \mathrm{kg}$, are attached to two identical massless springs. The spring-mass system is constrained to move inside a rigid smooth pipe bent in the form of a circle as shown in the figure. The pipe is fixed in a horizontal plane. The centres of the balls can move in a circle of radius $0.06\ \mathrm{m}$. Each spring has a natural length of $0.06\pi\ \mathrm{m}$ and spring constant $0.1\ \mathrm{N\ m^{-1}}$, Initially, both the balls are displaced by an angle $\theta = \frac{\pi}{6}\mathrm{rad}$ with respect to the diameter PQ of the circle (as shown in the figure) and released from rest. The frequency of oscillation of balls is



(A) $\frac{1}{2\pi}\mathrm{Hz}$

(B) $\frac{1}{3\pi}\mathrm{Hz}$

(C) $\frac{1}{\pi}\mathrm{Hz}$

(D) $\frac{1}{4\pi}\mathrm{Hz}$

Q.18

The refractive index of a prism for a monochromatic wave is $\sqrt{2}$ and its refracting angle is 60°. For minimum deviation, the angle of incidence will be

(A) 30°

(B) 45°

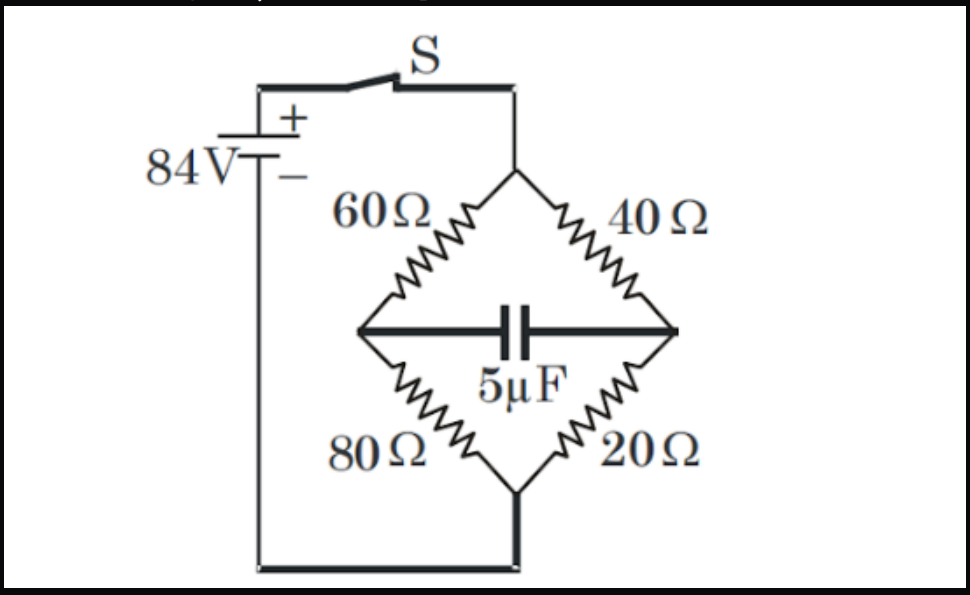
(C) 60°

(D) 75°

Q.19

The circuit shown in the figure is in steady state for a long time. The connection to battery is suddenly broken (switch S is opened up).

What is the charge (in μC) on the capacitor after 0.001 sec.?



(A) 100

(B) $100e^{-2}$

(C) $100e^{-4}$

(D) 0

Q.20

Which of these is correct match?

Misplaced \hline

(A) $C-2, D-1$

(B) $A-4,\ B-3$

(C) $A-3, C-2$

(D) $B-2,\ A-1$

Q.21

An installation consists of an electric motor which drives a water pump to lift 75 L of water per second to a height of 5 m, where water is disbursed at negligible speed. If the motor consumes a power of 5 kW, then what is the efficiency (%) of the installation? [$g = 10\ m\ s^{-2}$]

Q.22

The Poisson's ratio of a material is 0.4. If a force is applied to a wire of this material, there is a decrease of cross-sectional area by 2% . The percentage increase in its length is x. Then find the value of 10x.

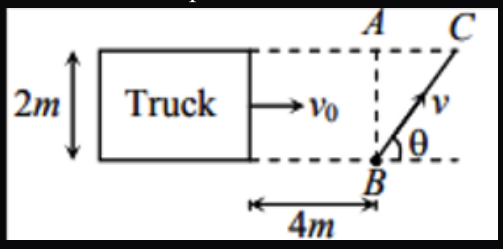
Q.23

The binding energy of an electron in the ground state of He is equal to 24.6 eV. The energy required to remove both the electrons is (mark answer to nearest integer)

Q.24

A 2 m wide truck is moving with a uniform speed $v_0 = 8$ m/s along a straight horizontal road. A pedestrian starts to cross the road

with a uniform speed v when the truck is $4\ m$ away from him. The minimum value of v so that he can cross the road safely is



(round off answer to nearest integer).

Q.25

A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in a time t. A number N of similar cells is now connected in series with a wire of the same material and cross-section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t. The value of N is

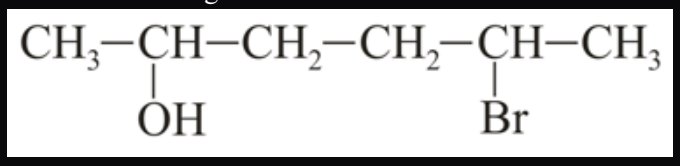
Q.26

Consider the following statement(s) for $CH_3 = X$ and $\dot{C}F_3 = Y'$
(I) When X dimerises, bond angle decreases
(II) When X dimerises, bond angle increases
(III) In $X - Y$ molecule, $C - C$ bond length is less than that in $Y - Y$ molecule
(IV) Bond angle in X is greater than that in Y. Pick the incorrect statements.

(A) II, III
(B) I, II, III
(C) I, IV
(D) II, III, IV

Q.27

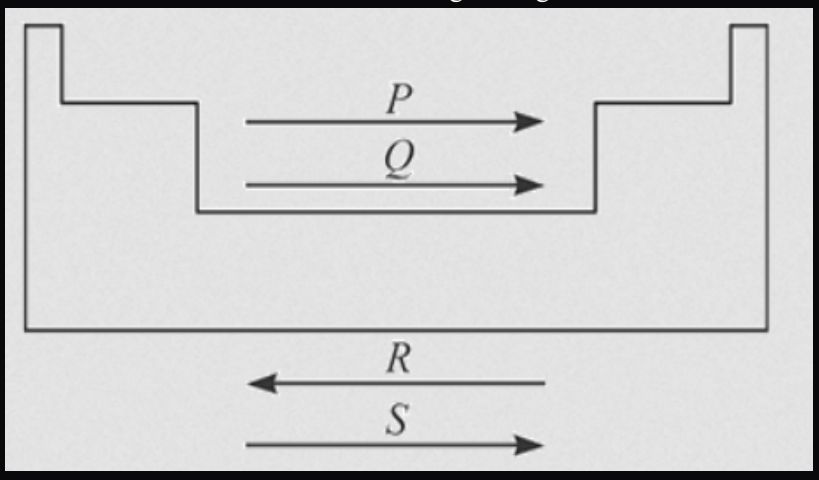
A mixture of all stereoisomers possible from the structure is subjected to fractional distillation. How many fractions will be obtained from the following?



(A) 2
(B) 4
(C) 6
(D) 7

Q.28

Choose the correct combination for the given figure.



(A)

P	Q	R	S
Ionisation enthalpy	Electron gain enthalpy	Atomic radius	Electro-negativity

(B)

P	Q	R	S
Atomic radius	Ionisation enthalpy	Electro-negativity	Electron gain

(C)

P	Q	R	S
Ionisation enthalpy	Atomic radius	Electro-negativity	Electron gain

(D)

P	Q	R	S
Electro-negativity	Electron gain enthalpy	Ionisation enthalpy	Atomic radius

Q.29

Find the final product ' E'

$$CH_2OH - CH = CH_2 \xrightarrow[\text{(ii) } NaSO_3H]{OsO_4} (A) \xrightarrow{KHSO_4} (B) \xrightarrow{Zn-Hg/\text{ conc. HCl}} (C) \xrightarrow{NBS/hv} (D) \xrightarrow{(CH_3)_2CuLi_4} (E)$$

(A) But-2-ene

(B) But-1-ene

(C) Isopentene

(D) Pen-2-ene

Q.30

Calcium lactate is a salt of weak acid i.e., lactic acid having general formula $Ca(LaC)_2$. Aqueous solution of salt has $0.3M$ concentration. pOH of solution is 5.60. If 90% of the salt is dissociated then what will be the value of pK_a ?

(A) $2.8 - \log(0.54)$

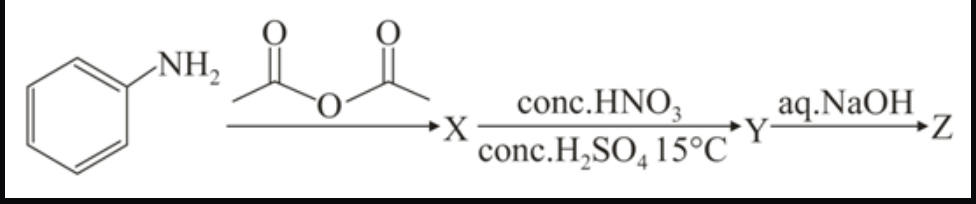
(B) $2.8 + \log(0.54)$

(C) $2.8 + \log(0.27)$

(D) None of these

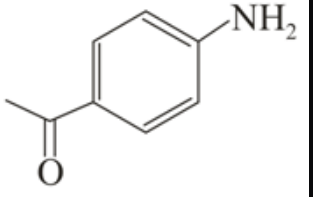
Q.31

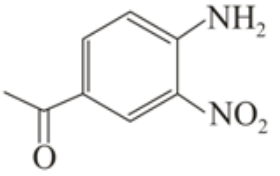
The major products X, Y and Z in the following sequence of transformations

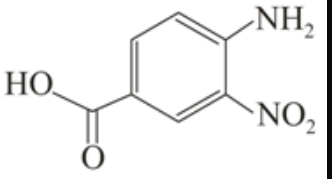


are:

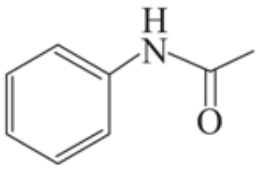
(A)

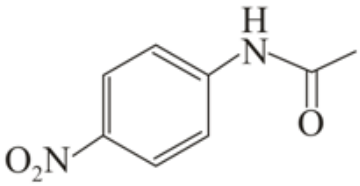
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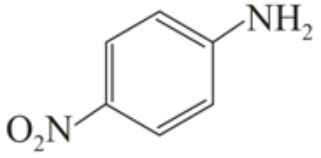
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Z = 

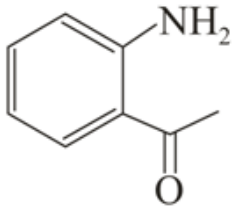
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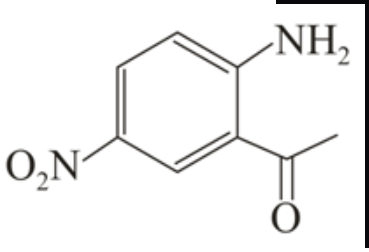
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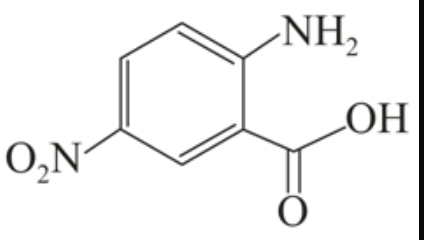
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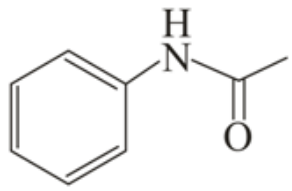
(C)

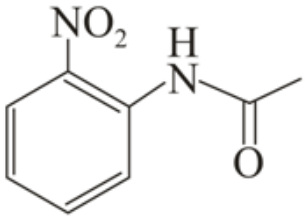
X = 

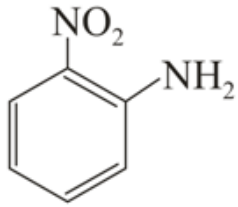
Y = 

Z = 

(D)

X = 

Y = 

Z = 

Q.32

Chlorine gas is prepared by reaction of H_2SO_4 with MnO_2 and $NaCl$. What volume of Cl_2 will be produced at STP if 50 g of $NaCl$ is taken in the reaction?

(A) 19.14 L

(B) 22.4 L

(C) 11.2 L

(D) 9.57 L

Q.33

The number of carbon atoms formed in the alkane, when a potassium salt of monocarboxylic acid containing n carbon atom undergoes Kolbe's electrolysis is:

(A) $2n$

(B) $2n + 2$

(C) $2n - 1$

(D) $2n - 2$

Q.34

The solutions of two compounds listed in Column-I when taken in molecular proportions give coordination compounds on evaporation. Match these compounds with their relative properties which is/are listed in Column-II.

Column-I	**Column-II**
A) $Fe(CN)_2 + 4KCN$	p) Coordination number of central metal ion in six
B) $CuSO_4 + 4NH_3$	q) Coordination number of central metal ion is four
C) $3NaF + AlF_3$	r) Coordination number of central metal ion is two
D) $AgCN + KCN$	s) Oxidation state of central metal ion is +2
	t) Overall charge on complex ion is +2

(A) A-p,s ; B-q,s,t; C-p; D-r

(B) A-q ; B-q,s,t; C-p; D-r

(C) A-s ; B-q,s,t; C-p; D-r

(D) A-p; B-q,s,t; C-p; D-r

Q.35

Enthalpy of fusion and enthalpy of vaporization for water respectively are $6.01\ \text{kJ mol}^{-1}$ and $45.07\ \text{kJ mol}^{-1}$ at 0°C what is enthalpy of sublimation at 0°C?

(A) $48.07\ \text{kJ mol}^{-1}$

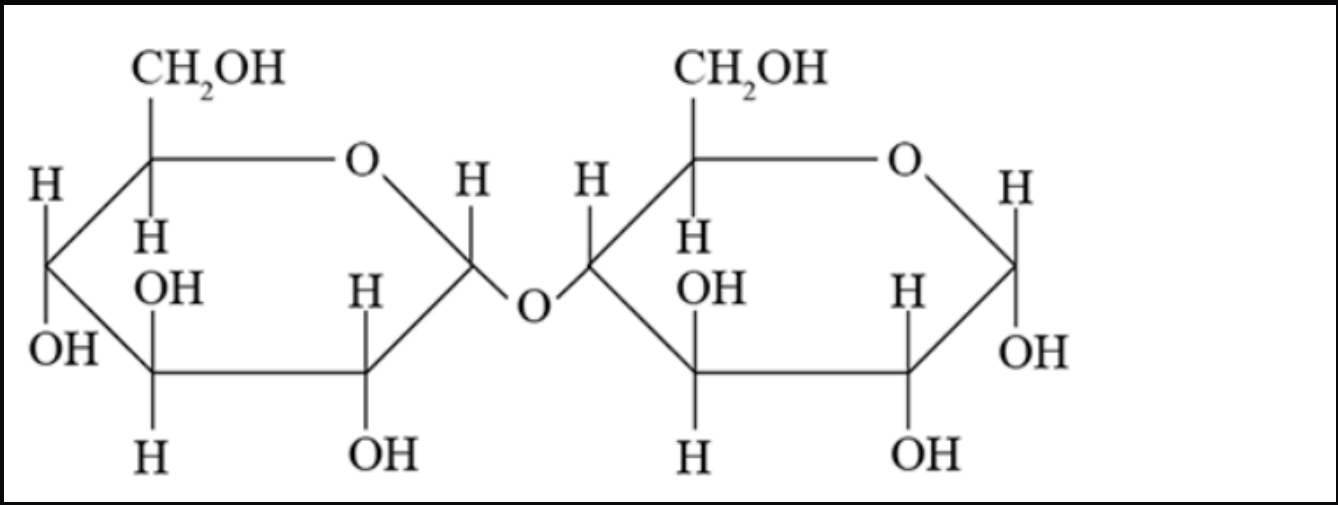
(B) $39.06\ \text{kJ mol}^{-1}$

(C) $51.08\ \text{kJ mol}^{-1}$

(D) $27.50\ \text{kJ mol}^{-1}$

Q.36

Study the structure of maltose and mark the incorrect statement.



(A) Maltose is composed of two α-D-glucose units

(B) $\text{C}-1$ of one glucose is linked to $\text{C}-4$ of other unit

(C) It is a non-reducing sugar

(D) It is a disaccharide

Q.37

Incorrect statement among the following is:

(A) Oxidation state of chromium in chromate and dichromate is same.

(B) Oxidation state of manganese is different in manganate and permanganate.

(C) Colour of chromate is yellow and dichromate is red.

(D) Chromate ion gets converted into dichromate ion in acidic medium.

Q.38

A metal gives two chlorides A and B. A gives black precipitate with NH_4OH and B gives white. With KI, B gives a red precipitate soluble in excess of KI. A and B are respectively :

(A) Hg_2Cl_2 and HgCl_2

(B) HgCl_2 and Hg_2Cl_2

(C) HgCl_2 and ZnCl_2

(D) ZnCl_2 and HgCl_{24}

Q.39

The correct order of boiling point is

(A) $T_2 < D_2 < H_2$

(B) n-pentane $<$ neo-pentane

(C) $Xe < Ar < He$

(D) m-nitrophenol $>$ o-nitrophenol

Q.40

What is the correct sequence of the increasing order of freezing points at one atmosphere of the following $1.0M$ aqueous solution?
1. Urea, 2. Sodium chloride, 3. Sodium sulphate, 4. Sodium phosphate.
Select the correct answer using the codes given below

(A) $4, 3, 1, 2$

(B) $3, 4, 2, 1$

(C) $3, 4, 1, 2$

(D) $4, 3, 2, 1$

Q.41

For a 3s-orbital, $\psi(3s) = \frac{1}{9\sqrt{3}}\left(\frac{1}{a_0}\right)^{3/2}\left(6 - 5\sigma + \sigma^2\right)e^{-\sigma/2}$where, $\sigma = \frac{2r.Z}{3a_0.}$
The maximum radial distance of node from nucleus will be

(A) $\frac{5a}{2Z}$

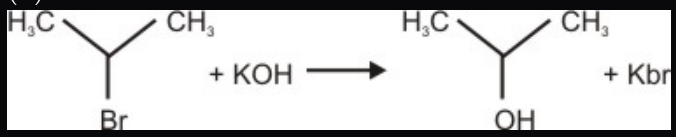
(B) $\frac{6a}{7Z}$

(C) $\frac{9a_0}{2Z}$

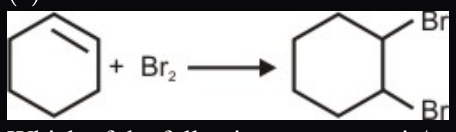
(D) $\frac{7a}{3Z}$

Q.42

For the following reactions:
(a) $CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH = CH_2 + KBr + H_2O$
(b)



(c)



Which of the following statements is/are CORRECT?

(A) (a) is elimination, (b) and (c) are substitution reactions

(B) (a) is substitution, (b) and (c) are addition reactions

(C) (a) and (b) are elimination reactions and (c) is addition reaction

(D) (a) is elimination, (b) is substitution and (c) is addition reaction

Q.43

Which among the following does not give Iodoform test?

(A) 1-Propanol

(B) Ethanol

(C) 2-Propanol

(D) Propanone

Q.44

A compound A dissociate by two parallel first order paths at certain temperature

$$\mathrm{A(g)} \overset{\mathrm{k_1(min^{-1})}}{\longrightarrow} 2\,\mathrm{B(\,g)} \mathrm{k_1} = 6.93 \times 10^{-3}\ \mathrm{min^{-1}}$$

$$\mathrm{A(\,g)} \overset{\mathrm{k_2(min^{-1})}}{\longrightarrow} \mathrm{C(g)} \mathrm{k_2} = 6.93 \times 10^{-3}\ \mathrm{min^{-1}}$$

If reaction started with pure 'A' with 1 mole of A in 1 litre closed container with initial pressure 2 atm. What is the pressure (in atm) developed in container after 50 minutes from start of experiment?

(A) 1.25

(B) 0.75

(C) 1.5

(D) 2.5

Q.45

The reaction of $\mathrm{RCONH_2}$ with a mixture of $\mathrm{Br_2}$ and aqueous KOH gives $\mathrm{RNH_2}$ as the main product. The intermediate not involved in this reaction is (are):

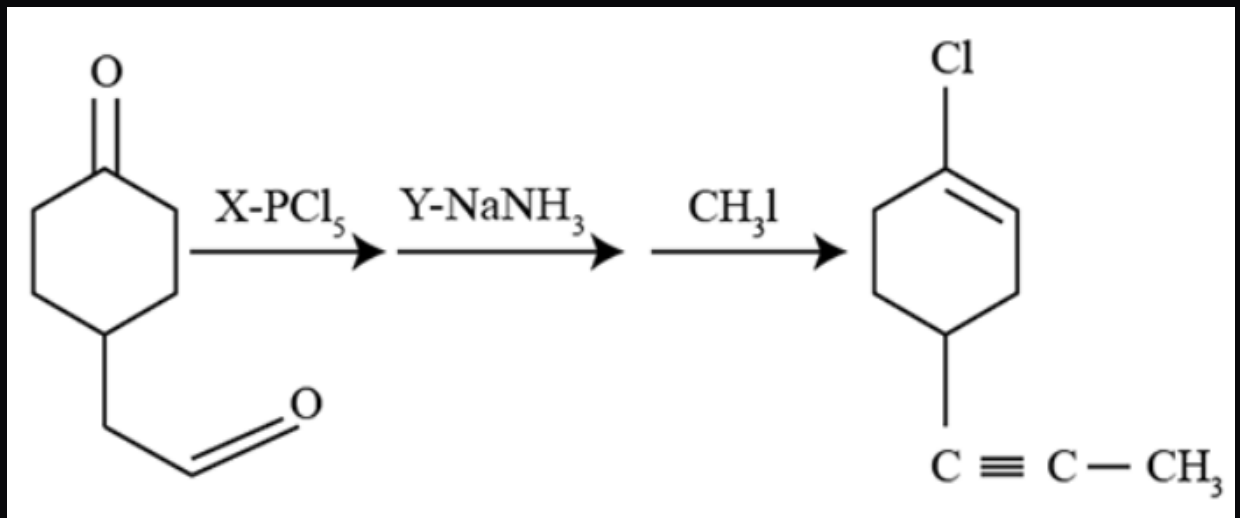
(A) $\mathrm{RCONHBr}$

(B) $\mathrm{RCON^-BrK^+}$

(C) $\mathrm{R-N=C=O}$

(D) $\mathrm{RCONBr_2}$

Q.46



The sum of $X + Y$ equal to

Q.47

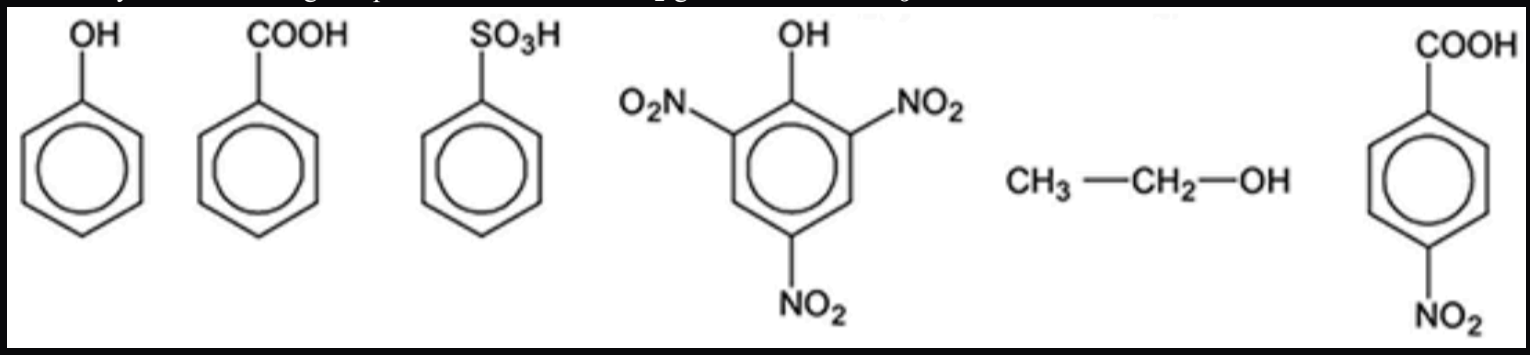
1 mole of carbon dioxide gas at $300\ \mathrm{K}$ is expanded under adiabatic condition such that its volume becomes 27 times. The work done during the process is $+\ldots\ldots$ cal (Given, $\gamma = 1.33$ and $C_V = 6\mathrm{calmol}^{-1}$ for CO_2)

Q.48

x g of a non-electrolytic compound (molar mass= 200) are dissolved in $1.0\ \mathrm{L}$ of $0.05\ \mathrm{M}$ NaCl aqueous solution. The osmotic pressure of this solution is found to be $4.92\ \mathrm{atm}$ at $27^{\circ}\mathrm{C}$. Calculate the value of x. Assume complete dissociation of NaCl and ideal behaviour of this solution.

Q.49

How many of the following compounds will liberate $\mathrm{CO_2}$ gas from $\mathrm{NaHCO_3}$ solution?



Q.50

Out of the following, how many oxides are amphoteric in nature?
$\mathrm{MnO},\ \mathrm{Mn_2O_3},\ \mathrm{MnO_2},\ \mathrm{MnO_3},\ \mathrm{Mn_2O_7},\ \mathrm{Mn_3O_4}$

Q.51

A relation R is defined as $(x,\ y) \in R \Rightarrow x^y = y^x$ for $x,\ y \in I - \{0\}$, where I is the set of all integers. Then the relation R is:

(A) reflexive but not symmetric

(B) svmmetric but not reflexive

(C) reflexive and symmetric both

(D) equivalence relation

Q.52

Let $f(x) = \begin{cases} x^3 + x^2 - 10x & , \quad -2 \le x < 0 \\ \sin x & , \quad 0 \le x < \frac{\pi}{2}, \text{ then } f(x) \\ 1 + \cos x & , \quad \frac{\pi}{2} \le x < \pi \end{cases}$

(A) has a local maxima at $x = 0$ and $x = \frac{\pi}{2}$

(B) has a local minima at $x = \frac{\pi}{2}$

(C) has a local minima at $x = \frac{\pi}{2}$ and a local maxima at $x = 0$

(D) has a local minima at $x = 0$ and a local maxima at $x = \frac{\pi}{2}$

Q.53

If the line $y = x\tan\theta$ cuts the curve $x^3 + xy^2 + 2x^2 + 2y^2 + 3x + 1 = 0$ at the points A, B and C. If OA, OB and OC are in HP, then $\tan\theta$ is equal to

(A) ± 1

(B) 0

(C) 2

(D) -2

Q.54

If two different numbers are taken from the set $\{0, 1, 2, 3, \ldots, 10\}$, then the probability that their sum as well as absolute difference are both multiple of 3 is:

(A) $\frac{12}{55}$

(B) $\frac{14}{45}$

(C) $\frac{7}{55}$

(D) $\frac{6}{55}$

Q.55

Let $f(x)$ be a polynomial satisfying $f(x)f\left(\frac{1}{x}\right) + 5 - 3f(x) - 3f\left(\frac{1}{x}\right) = 0, \forall x \in R - \{0\}$ and $f(2) = 11$, then $f(3)$ is equal to:

(A) 21

(B) 12

(C) 20

(D) 11

Q.56

Let $\{\Delta_1, \Delta_2, \Delta_3, \ldots\ldots, \Delta_k\}$ be the set of all third-order determinants that can be made with the distinct non-zero real numbers $a_1,\ a_2,\ a_3, \ldots\ldots\ldots, a_9$. Then

(A) $k = 8!$

(B) $\sum\limits_{i=1}^{k} \Delta_i = 0$

(C) At most one $\Delta_i = 0$

(D) None of these

Q.57

Let $\mathrm{I}_1 = \int\limits_0^1 \frac{|\ln x|}{x^2+4x+1}dx$ and $\mathrm{I}_2 = \int\limits_1^\infty \frac{\ln x}{x^2+4x+1}dx$, then

(A) $\mathrm{I}_1 = \mathrm{I}_2$

(B) $\mathrm{I}_1 > \mathrm{I}_2$

(C) $\mathrm{I}_1 + \mathrm{I}_2 = 0$

(D) $\mathrm{I}_1 = 2\mathrm{I}_2$

Q.58

The coefficient of variation of two series are 70 and 90 and their standard deviation are 17. 5 and 18, respectively. The mean of two series are

(A) $25,\ 20$

(B) $18,\ 22$

(C) $22,\ 18$

(D) $16,\ 24$

Q.59

If the straight lines $\frac{x}{1}=\frac{y}{1}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ and a third line passing through the point $Q(1,1,1)$ form a triangle whose area is $6\frac{1}{2}$ sq. units, then the point of intersection of second line with third line is:

(A) $(1,\ 2,\ 3)$

(B) $(2,\ 4,\ 6)$

(C) $\left(\frac{4}{3},\ \frac{8}{3},\ \frac{12}{3}\right)$

(D) $(2,\ 1,\ 3)$

Q.60

The function f(x) = {x} sin (π[x]), where [.] denotes the greatest integer function and {.} is the fractional part function, is discontinuous at

(A) all x

(B) all integer points

(C) no x

(D) x which is not an integer

Q.61

Let the position vectors of points $A,\ B$ and C in ΔABC be $\hat{i}+\hat{j}+2\widehat{k},\ \hat{i}+2\hat{j}+\widehat{k}$ and $2\hat{i}+\hat{j}+\widehat{k}$, respectively. Let $l_1,\ l_2$ and l_3 be the lengths of the perpendiculars drawn from the orthocentre O on the sides $AB,\ BC$ and CA respectively. Then $(l_1+l_2+l_3)$ is equal to

(A) $\frac{4}{\sqrt{6}}$

(B) $\frac{\sqrt{6}}{2}$

(C) $\frac{3}{\sqrt{6}}$

(D) $\frac{\sqrt{6}}{5}$

Q.62

The area above the x-axis enclosed by the curves $x^2-y^2=0$ and $x^2+y-2=0$ is

(A) $\frac{5}{3}$

(B) $\frac{7}{3}$

(C) $\frac{8}{3}$

(D) $\frac{10}{3}$

Q.63

If P is an orthogonal matrix and $Q = PAP^T$ and $B = P^T Q^{1000} P$, then B^{-1} is, where A is involutory matrix

(A) A

(B) A^{1000}

(C) I

(D) None of these

Q.64

All the words formed by writing all the letters of word ZENITH are arranged as in English dictionary. Now the position of word ZENITH is (from beginning)

(A) 598

(B) 602

(C) 532

(D) 616

Q.65

Solution of the differential equation,
$xdy = \frac{xydx}{\sqrt{1-x^2}} - ydx$ is (where $x \in (-1, 1)$)

(A) $\log_e |xy| = \sin^{-1} x + C$

(B) $\log_e |xy| = 2\sin^{-1} x + C$

(C) $\log_e |(x+y)| = \sin^{-1} x + C$

(D) $\log_e |(x+y)| = 2\sin^{-1} x + C$

Q.66

If $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$, then the vector $\vec{r}$ satisfying $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} . \vec{a} = 0$ is

(A) $\hat{i} + 8\hat{j} + 2\hat{k}$

(B) $\hat{i} - 8\hat{j} + 2\hat{k}$

(C) $\hat{i} - 8\hat{j} - 2\hat{k}$

(D) $-\hat{i} - 8\hat{j} + 2\hat{k}$

Q.67

If z_1, z_2, z_3 are the roots of the equation $x^3 + 3px^2 + 3qx + r = 0$ (where p, q, r are also complex numbers) correspond to points A, B and C. If triangle is ΔABC is equilatral then which of the following is correct ?

(A) $p = q^2$

(B) $p^2 = 3q$

(C) $p^2 = q$

(D) $q^2 = 3p$

Q.68

If $a_1,\ a_2,\ a_3, \ldots, a_{20}$ are the arithmetic means between 13 and 67, then the maximum value of the product $a_1 \cdot a_2 \cdot a_3 \cdot \ldots a_{20}$ is

(A) $(20)^{20}$

(B) $(60)^{20}$

(C) $(80)^{20}$

(D) $(40)^{20}$

Q.69

From the focus $(-5,\ 0)$ of the ellipse $\frac{x^2}{45} + \frac{y^2}{20} = 1$, a ray of light is sent which makes an angle $\cos^{-1}\left(\frac{-1}{\sqrt{5}}\right)$ with the positive direction of x-axis, upon reaching the ellipse surface, the ray is reflected from it. Slope of the reflected ray is

(A) $-\frac{3}{2}$

(B) $-\frac{7}{3}$

(C) $-\frac{5}{4}$

(D) $-\frac{2}{11}$

Q.70

Given two real sets $A = \{a, a_2, a_3 \cdots a_{2n}\}$ and $B\{b_1, b_2, \cdots b_n\}$. If $f : A \to B$ is a function such that every element of B has an inverse image and $f(a_1) \leq f(a_2) \leq f(a_3) \leq f(a_4) \cdots \leq f(a_{2n})$, then the number of such mappings are

(A) ${}^{2n}C_n$

(B) ${}^{2n}C_{n-1}$

(C) ${}^{2n-1}C_{n-1}$

(D) ${}^{2n+1}C_n$

Q.71

Two rods of length a and b slide along the axes in a manner that their ends are always concyclic. If the locus of the centre of the circle passing through these ends is $k\left(x^2 - y^2\right) = a^2 - b^2$, then find the value of k.

Q.72

If α and β are the roots of the equation $x^2 - 2x + 3 = 0$, then the sum of roots of the equation having roots as $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ and $\beta^3 - \beta^2 + \beta + 5$ is

Q.73

n is selected from the set $\{1, 2, 3, \ldots\ldots\ldots 100\}$ and the number $2^n + 3^n + 5^n$ is formed. Let the total number of ways of selecting n is P, so that the formed number is divisible by 4, then $\frac{P}{7}$.

Q.74

Let A and B are 3×3 matrices with real number entries, where A is symmetric, B is skew-symmetric and $(A + B)(A - B) = (A - B)(A + B)$. If $(AB)^T = (-1)^k AB$, then the sum of all possible integral value of k in $[2, 10]$ is equal to (where A^T represent transpose of matrix A)

Q.75

Number of integral values of k for which the equation $4\cos^{-1}(-|x|) = k$ has exactly two solutions, is

Answers & Solutions

Q.1 Answer:

$\frac{GM}{2R^2}$

Solution:

By the principle of superposition of fields

$\overrightarrow{\mathrm{E}} = \overrightarrow{\mathrm{E}}_1 + \overrightarrow{\mathrm{E}}_2$

Here, $\overrightarrow{\mathrm{E}}$ **= net field at the centre of hole due to entire mass** $\overrightarrow{\mathrm{E}}_1$ **= field due to remaining mass and** $\overrightarrow{\mathrm{E}}_2$ **= field due to mass in hole** $= 0$

$\therefore \mathrm{E}_1 = \mathrm{E} = \left(\frac{\mathrm{GM}}{\mathrm{R}^3}\right) \cdot \mathrm{r}$ **where** $\mathrm{r} = \frac{\mathrm{R}}{2}$

$\therefore \mathrm{E} = \frac{\mathrm{GM}}{2\mathrm{R}^2}$

Q.2 Answer:

Force is required and heat is dissipated in the sheet.

Solution:

Heat will develop due to eddy current which will oppose the motion and hence a force is required.

Q.3 Answer:

H_2

Solution:

As, $m = \frac{3kT}{c^2} = \frac{3RT}{Nc^2}$

$= \frac{3\times 8.31\times 300}{6.023\times 10^{23}\times (1930)^2} = 3.3\times 10^{-27}\ \mathrm{kg}$

Mass of H_2 **molecule** $= 1.66\times 10^{-27}\times 2$

$= 3.32\times 10^{-27}\ \mathrm{kg}$

Thus, the gas is hydrogen.

Q.4 Answer:

$9\times 10^{13}\ \mathrm{J}$

Solution:

The amount of mass converted into heat will be 0.1% **of**

$1\ \mathrm{kg} = 1/1000\mathrm{Kg}$

Now energy released $= \Delta\mathrm{E} = \mathrm{mc}^2 = 1/1000\times\left(3\times 10^8\right)^2$

$= 9\times 10^{16-3}$

$= 9\times 10^{13}\ \mathrm{J}$

Q.5 Answer:

$n^{1/3} : 1$

Solution:

$n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow R = r(n)^{1/3}$

$C_{\text{Big}} = (n)^{1/3} C_{\text{small}}$

$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2; \quad V = \frac{kq}{r} = \frac{q}{c}$

$U \propto c$ when V is same

$U \propto n^{1/3}$

if q same, $U \propto \frac{1}{n^{1/3}}$

Q.6 Answer:

Assertion is false but Reason is true.

Solution:

Let energy corresponding to the blue light is equal to the work function of the metal. If the frequency of incident radiation is greater than the threshold frequency, i.e., greater than the frequency of blue light for the given surface, a photoelectron can be emitted from it.

However, since red light has a lower frequency than blue light, it cannot eject photoelectron from the given surface.

Violet light has greater energy than blue light, while red light has smaller energy than blue light. As a result, violet light emits photoelectron while red light does not.

Now, the expression of the energy of the photon is $E = h\nu$, here ν is the frequency and h is Planck's constant.

Thus, from the above relation, we get, $E \propto \nu$.

Since violet light has a higher frequency than red light. As a result, violet light photons are more energetic than red light photons.

Q.7 Answer:

9×10^5

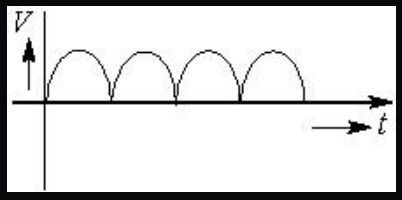
Solution:

$$\frac{\mathrm{P}}{\mathrm{Q}} = \frac{\mathrm{R}}{\mathrm{X}} \quad \mathrm{X}_{\max} = \frac{\mathrm{R}_{\max} \cdot \mathrm{Q}_{\max}}{\mathrm{P}_{\min}}$$

$$= 9000 \times \frac{1000}{10} = 9 \times 10^5 \text{ohm}$$

$$\mathrm{X}_{\min} = \frac{\mathrm{R}_{\min} \cdot \mathrm{Q}_{\min}}{\mathrm{P}_{\max}} = 100 \times \frac{10}{1000} = 1\text{ohm}$$

Q.8 Answer:



Solution:

Speed can never be negative. Hence (2) is correct.

Q.9 Answer:

$\frac{\mu_0}{2\pi}$

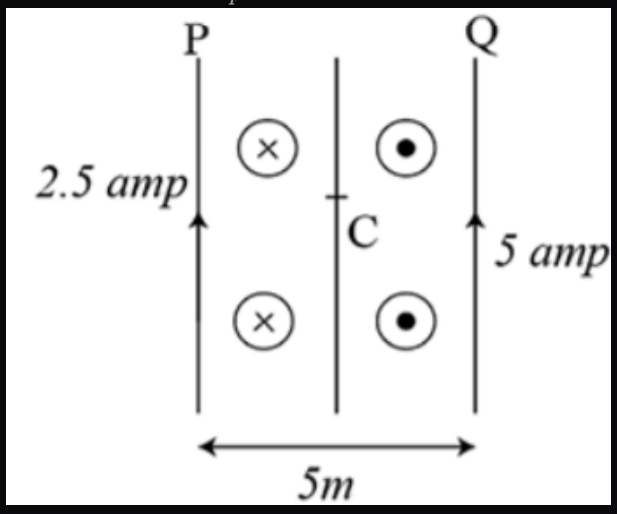
Solution:

When the current flows in both wires in the same direction then magnetic field at half way due to the wire P,

$\overrightarrow{\mathrm{B}}_{\mathrm{p}}=\frac{\mu_0 \mathrm{I}_1}{2 \pi \frac{5}{2}}=\frac{\mu_0 \mathrm{I}_1}{\pi \cdot 5}=\frac{\mu_0}{2 \pi}$

(where $I_1=5$ amp)

The direction of $\vec{B}_p$ is downward $\odot$



Magnetic field at half way due to wire Q

$\overrightarrow{\mathrm{B}}_{\mathrm{Q}}=\frac{\mu_0 \mathrm{I}_2}{2 \pi \frac{5}{2}}=\frac{\mu_0}{\pi}$ (upward)

[where $\mathrm{I}_2=5$ amp.]

Net magnetic field at half way

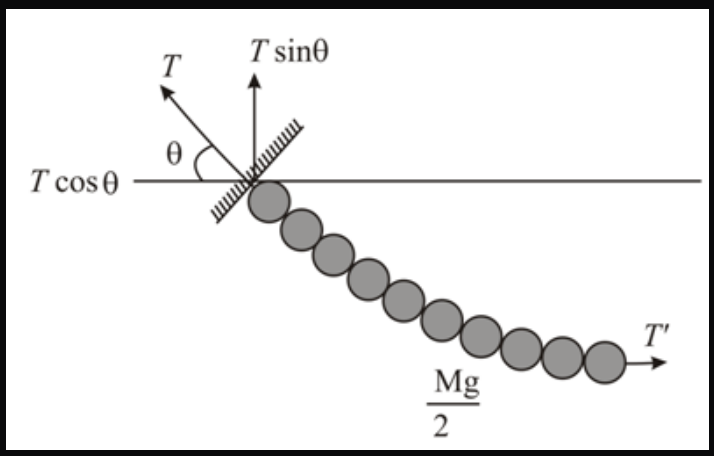
$\overrightarrow{\mathrm{B}}=\overrightarrow{\mathrm{B}}_{\mathrm{P}}+\overrightarrow{\mathrm{B}}_{\mathrm{Q}}=-\frac{\mu_0}{2 \pi}+\frac{\mu_0}{\pi}=\frac{\mu_0}{2 \pi}$ (upward)

Hence, net magnetic field at midpoint $=\frac{\mu_0}{2 \pi}$

Q.10 Answer:

$\frac{Mg\cot\theta}{2}$

Solution:



Now from the above free body diagram,

$T\sin\theta=\frac{Mg}{2}$,

Also, let $T\prime$ be the horizontal component such that,

$T\cos\theta=T\prime$,

Now using the above two equations,

$\frac{T\sin\theta}{T\cos\theta}=\frac{\frac{Mg}{2}}{T\prime}$

Now solving the above equation we get,

$\frac{1}{\cot\theta}=\frac{\frac{Mg}{2}}{T\prime}\Rightarrow T\prime=\frac{Mg\cot\theta}{2}$

Q.11 Answer:

$2I_0$

Solution:

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

Given, $l_1 = I_0, I_2 = 4/_0$

Then,

$I_A = I_0 + 4I_0 + 2 \times \sqrt{I_0 \times 4I_0} \times \cos\frac{\pi}{3}$

$= 5l_0 + 2I_0 = 7I_0$

$= 5l_0 + 2I_0 = 7I_0$

and

$I_B = I_0 + 4I_0 + 2\sqrt{4I_0^2} \times \cos\pi/2$

$= 5I_0 + 0 = 5I_0$

$\therefore I_A - I_B = 7I_0 - 5I_0 = 2I_0$

$= 5I_0 + 0 = 5I_0$

$\therefore I_A - I_B = 7I_0 - 5I_0 = 2I_0$

Q.12 Answer:

2.8 mm

Solution:

Change in length of brass rod

$\Delta l_B = \alpha_B l_B (T_2 - T_1)$

$= 2.5 \times 10^{-5} \times 500 \times (200 - 50)$

= 1.875 mm

Similarly change in length of the steel rod

$\Delta l_s = \alpha_s l_S (T_2 - T_1)$

$= 1.25 \times 10^{-5} \times 500 \times (200 - 50)$

= 0.9375 mm

Therefore, change in length of the combined rod

$= \Delta l_B + \Delta l_S = 1.875 + 0.9375$

= 2.8175 mm =2.8 mm

Q.13 Answer:

$1.75\ \text{kg m}^{-3}$

Solution:

If M **is molecular mass of the gas, then from**

$M\,(C_p - C_v) = R$

$M = \frac{8.31}{210} = 0.0392$

If ρ **is density of the gas at NTP, then mass of** $1\ \text{m}^3$ **of gas at** $\text{NTP} = \rho\text{kg}$

$\therefore$ **Mass of** $22.4\ \text{L}\ (= 22.4 \times 10^{-3}\ \text{m}^3)$ **of gas at** $\text{NTP} = \rho \times 22.4 \times 10^{-3}\ \text{kg}$**, which is the molecular mass of the gas**

$\therefore \rho \times 22.4 \times 10^{-3} = 0.0392$

$\rho = \frac{0.0392}{22.4 \times 10^{-3}} = 1.75\ \text{kg m}^{-3}$

Q.14 Answer:

an OR gate and an AND gate respectively

Solution:

For first case, the Boolean expression is, $Y = \overline{\bar{A} \cdot \bar{B}} = A + B$ hence for OR gate and for second case, the Boolean expression is $Y = \overline{\bar{A} \cdot \bar{B}} = A.B$, and hence for AND gate.

Q.15 Answer:

$0.08\mathrm{H}$

Solution:

A coil consists of an inductance (L) and a resistance (R)

In dc only resistance is effective. Hence, $\mathrm{R} = \frac{\mathrm{V}}{\mathrm{i}} = \frac{12}{4} = 3\Omega$

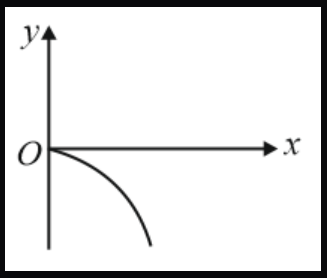
In ac, $i_{\mathrm{rms}} = \frac{\mathrm{V_{rms}}}{\mathrm{Z}} = \frac{\mathrm{V_{rms}}}{\sqrt{\mathrm{R}^2 + w^2\,\mathrm{L}^2}}$

$\therefore \mathrm{L}^2 = \frac{1}{\omega^2}\left[\left(\frac{\mathrm{V_{rms}}}{\mathrm{i_{rms}}}\right)^2 - \mathrm{R}^2\right]$

$\Rightarrow \mathrm{L} = \frac{1}{\omega}\sqrt{\left(\frac{\mathrm{V_{rms}}}{\mathrm{i_{rms}}}\right)^2 - \mathrm{R}^2} = \frac{1}{50}\sqrt{\left(\frac{12}{2.4}\right)^2 - (3)^2}$

$= 0.08$ **henry**

Q.16 Answer:



Solution:

Path of the centre of mass in a two particle system,

$r(t) = \left[\frac{m_1 r_1(t) + m_2 r_2(t)}{m_1 + m_2}\right]$

or $r(t) = \left[\frac{m\left(t\,\hat{\mathrm{i}} - t^3\,\hat{\mathrm{j}} + 2t^2\,\hat{\mathrm{k}}\right) + 2m\left(t\,\hat{\mathrm{i}} - t^3\,\hat{\mathrm{j}} - t^2\,\hat{\mathrm{k}}\right)}{m + 2m}\right]$

$\Rightarrow \quad r(t) = \frac{3t\,\hat{\mathrm{i}} - 3t^3\,\hat{\mathrm{j}}}{3} = t\,\hat{\mathrm{i}} - t^3\,\hat{\mathrm{j}}$

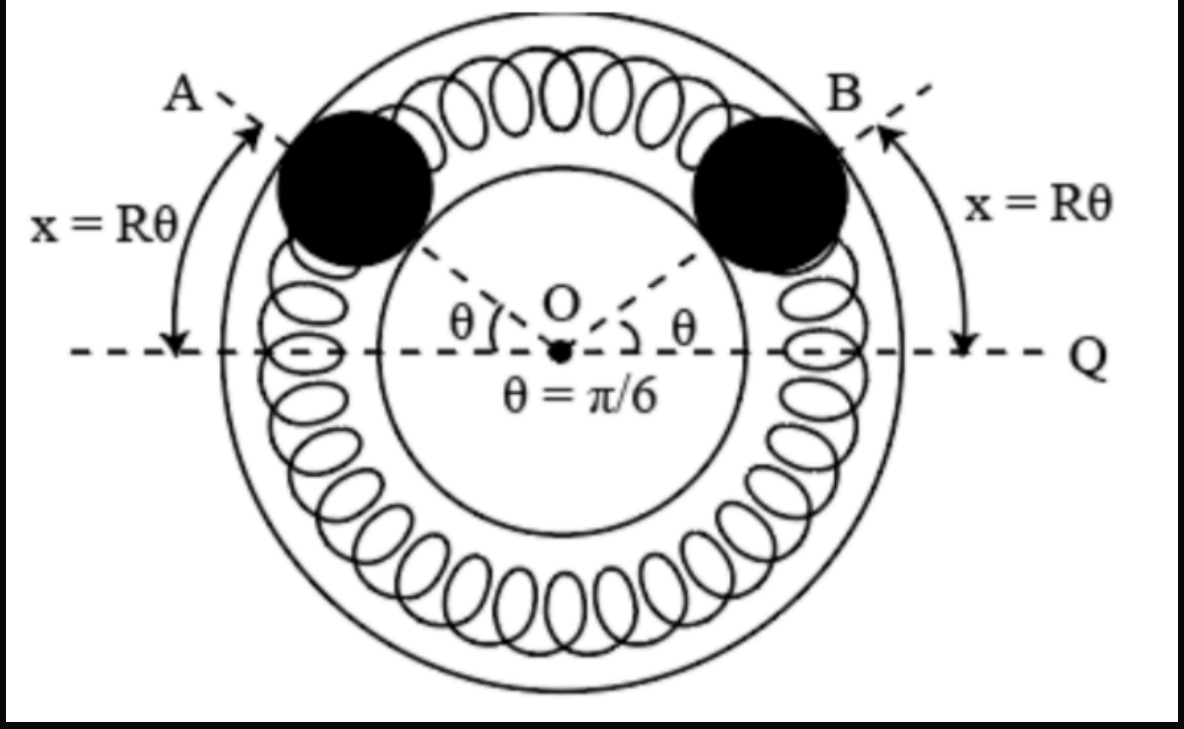
Q.17 Answer:

$\frac{1}{\pi}\mathrm{Hz}$

Solution:

Given, Mass of each block A and B, $m = 0.1\ \mathrm{kg}$

Radius of circle, $R = 0.06\ \mathrm{m}$

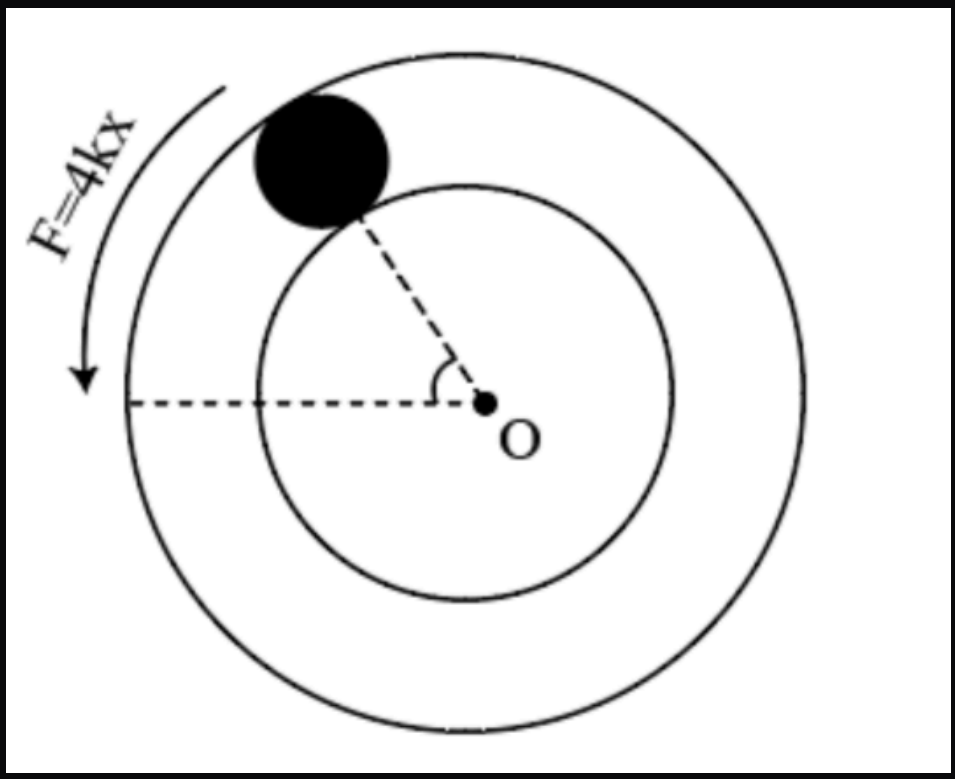


The natural length of spring $l_0 = 0.06\pi = \pi \mathrm{R}$ (Half circle) and spring constant, $\mathrm{k} = 0.1\ \mathrm{N\ m^{-1}}$

In the stretched position elongation in each spring

$\mathrm{x} = \mathrm{R}\theta$

Let us draw FBD of A



Spring in lower side is stretched by $2x$ and on upper side compressed by $2x$.

Therefore, a force $2\mathrm{kx}$ on each block would be exerted by each spring.

Hence, a restoring force, $\mathrm{F} = 4\mathrm{kx}$ will act on A in the direction shown in figure.

Restoring torque of this force about origin

$\tau = -\mathrm{F} \cdot \mathrm{R} = -(4\mathrm{kx})\mathrm{R} = -(4\mathrm{kR}\theta)\mathrm{R}$

or $\tau = -4kR^2 \cdot \theta \quad \ldots\ldots$ **(i)**

Since, $\tau \propto -\theta$, each ball executes angular SHM about origin O.

Equation. (i) can be written as

$\mathrm{I}\alpha = -4\mathrm{kR}^2\theta$

or $\left(\mathrm{mR}^2\right)\alpha = -4\mathrm{kR}^2\theta$

or $\alpha = -\left(\frac{4k}{m}\right)\theta$

Frequency of oscillation, $f = \frac{1}{2\pi}\sqrt{\frac{\text{acceleration}}{\text{displacement}}}$

$= \frac{1}{2\pi}\sqrt{\left|\frac{\alpha}{\theta}\right|}$

$f = \frac{1}{2\pi}\sqrt{\frac{4k}{m}}$

Substituting the values, we have

$f = \frac{1}{2\pi}\sqrt{\frac{4\times0.1}{0.1}} = \frac{1}{\pi}Hz$

Q.18 Answer:

45°

Solution:

As, $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$

$\Rightarrow \quad \sqrt{2} = \frac{\sin\left(\frac{60^\circ+\delta_m}{2}\right)}{\sin\frac{60^\circ}{2}} (\text{as } \mu = \sqrt{2})$

$\frac{1}{\sqrt{2}} = \sin\left(\frac{60^\circ+\delta_m}{2}\right)$

or $\sin 45^\circ = \sin\left(\frac{60^\circ+\delta_m}{2}\right)$

$\Rightarrow \delta_m = 30^\circ$

$\therefore$ Angle of incidence,

$i = \frac{A+\delta_m}{2} = \frac{60+30}{2} = \frac{90}{2} = 45^\circ$

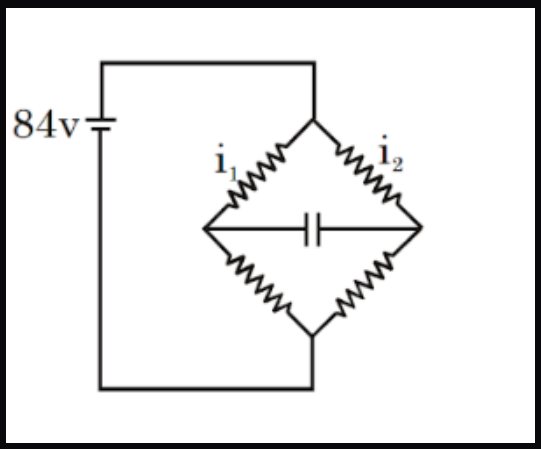
Q.19 Answer:

$100e^{-4}$

Solution:

$i_1 = \frac{84}{140} = 0.6\ A$

$i_2 = \frac{84}{60} = 1.4\ A$

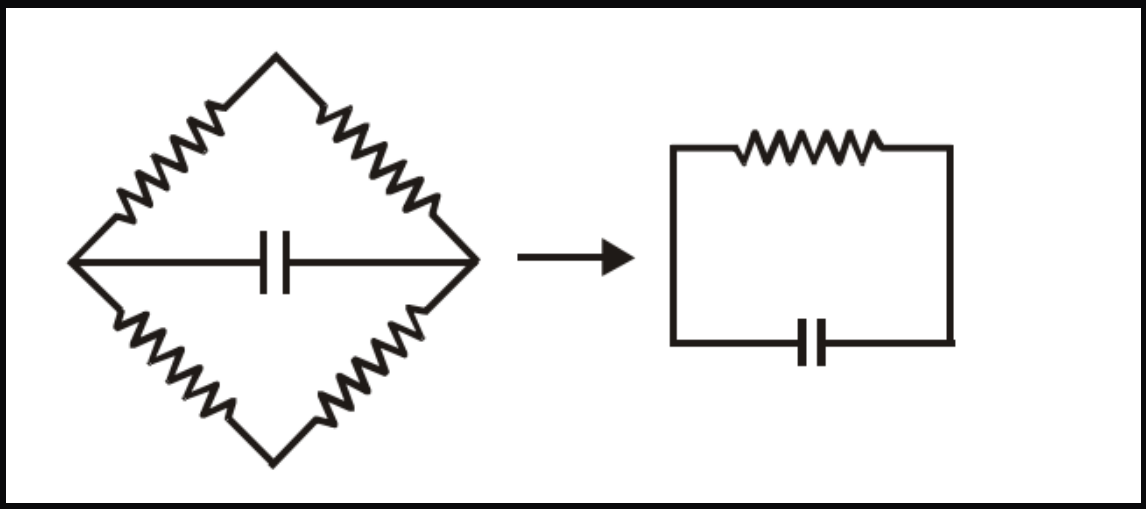


p.d across capacitor $= 20v$

$Q = CV = 5 \times 20 = 100\mu C$ **After** S **is open**

$Q = Q \cdot e^{-t/RC}$

$= 100e^{-4}$



Q.20 Answer:

$A - 4,\ B - 3$

Solution:

Dimension of torque $= \left[M^1\ L^2\ T^{-2}\right]$

Dimension of angular momentum $= \left[M^1\ L^2\ T^{-1}\right]$

$\therefore$ Relation $A - 4,\ B - 3$ **is right.**

Q.21 Answer:

—

Solution:

Power consumed by the motor

$P_{motor} = 5000\ W$

Power used in lifting the water

$P_{water} = \frac{mgh}{t} = \frac{75 \times 10 \times 5}{1} = 3750\ W$

The efficiency of the motor is

$\eta = \frac{P_{water}}{P_{motor}} \times 100\%$

$\eta = \frac{3750}{5000} \times 100\% = 75\%$

Q.22 Answer:

—

Solution:

Poisson's ratio is defined as $\sigma = \frac{\Delta d/d}{\Delta l/l}$ **or** $\frac{\Delta l}{l} = \frac{\Delta d/d}{\sigma}$ **... (i)**

where d **is the diameter and** l **is the length of the wire. The area of cross-section is**

Or $A = \pi r^2 = \frac{\pi d^2}{4}$

$\log A = \log\left(\frac{\pi}{4}\right) + 2\log d$

Differentiating, we have

Or $\frac{\Delta A}{A} = 2\frac{\Delta d}{d}$

$\frac{\Delta d}{d} = \frac{1}{2}\frac{\Delta A}{A} = \frac{1}{2} \times 2\% = 1\%$

Using this in

(i) we have $\left(\because \frac{\Delta A}{A}=2\%, \text{given }\right)$
$\frac{\Delta l}{l}=\frac{1\%}{0.4}=2.5\% \quad (\because \sigma=0.4)$

Q.23 Answer:

—

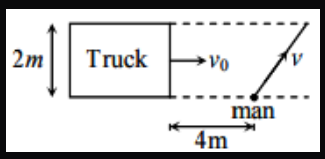
Solution:

Helium atom has 2 electrons. When one electron is removed, the remaining atom is hydrogen like atom, whose energy in first orbit is
$E_1=-(2)^2(13.6 \text{ eV})=-54.4 \text{ eV}$
Therefore, to remove the second electron from the atom, the additional energy of 54.4 eV is required. Hence, total energy required to remove both the electrons
$=24.6+54.4=79.0 \text{ eV}$

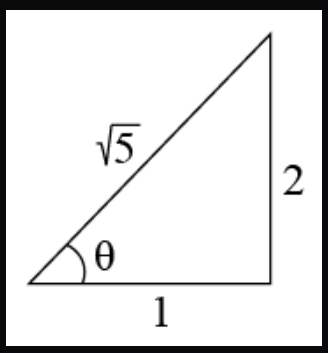
Q.24 Answer:

—

Solution:



Let the man starts crossing the road at an angle θ as shown in figure. For safe crossing the condition is that the man must cross the road by the time the truck describes the distance $4+\text{AC}$ or $4+2\cot\theta$.
$\therefore \frac{4+2\cot\theta}{8}=\frac{2/\sin\theta}{v}$
or $v=\frac{8}{2\sin\theta+\cos\theta} \ldots (1)$
For minimum v, $\frac{dv}{d\theta}=0$
or $\frac{-8(2\cos\theta-\sin\theta)}{(2\sin\theta+\cos\theta)^2}=0$
or $\quad 2\cos\theta-\sin\theta=0$
or $\quad \tan\theta=2$



From Eq. (1)
$v_{\min}=\frac{8}{2\left(\frac{2}{\sqrt{5}}\right)+\frac{1}{\sqrt{5}}}=\frac{8}{\sqrt{5}}=3.57 \text{m/s}$

Q.25 Answer:

—

Solution:

As, $R=\frac{\rho L}{A}; R_1=\frac{\rho 2L}{A}=2R$
$m=ALd; m_1=A2Ld=2m$

where d is density.

Now, $\frac{(3V)^2t}{R} = mc\Delta T \quad \ldots(i)$

and $\frac{(\mathrm{NV})^2\mathrm{t}}{2R} = 2mc\Delta T \quad \ldots(ii)$

Solving Eqs. (i) and (ii), we get $N = 6$

Q.26 Answer:

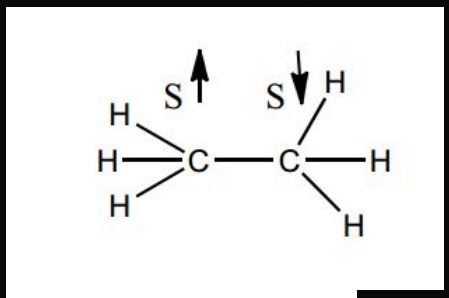
II, III

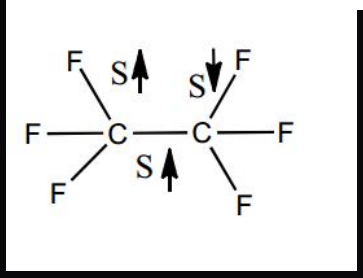
Solution:

In CH_3 bond angle, $\approx 120^\circ \left(\mathrm{sp}^2 - \text{ hybridisation }\right)$

In $\dot{C}F_3$ bond angle, $< 109^\circ \left(\mathrm{sp}^3 - \text{ hybridisation }\right)$

In C_2H_6 , carbon is sp^3 hybrid, so, bond angle decreases to $\approx 109^\circ$.





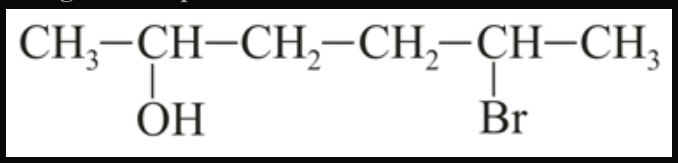
So, the bond length decreases in C_2F_6.

Q.27 Answer:

2

Solution:

The given compound is:



The stereoisomers of the given compound are:

$\left.\begin{matrix}\text{I } 2\mathrm{R},\ 5\mathrm{R} \\ \text{II } 2\mathrm{S},\ 5\mathrm{S}\end{matrix}\right\}$ represent $(\pm)$ pair $\left.\begin{matrix}\text{III } 2\mathrm{R},\ 5\mathrm{S} \\ \text{IV } 2\mathrm{S},\ 5\mathrm{R}\end{matrix}\right\}$ represent $(\pm)$ pair

On fractional distillation of the mixture, we get only two fractions (of two enantiomers $(\pm)$ pair) because enantiomers have the same Physical properties.

Q.28 Answer:

P	Q	R	S
Ionisation enthalpy	Electron gain enthalpy	Atomic radius	Electro-negativity

Solution:

Ionisation enthalpy, electron gain enthalpy and electronegativity increases in a period from left to right while atomic radius decreases.

Q.29 Answer:

But-1-ene

Solution:

Here the final product E is Butene and the reaction sequence is as follows

$$\underset{}{CH_2OH-CH=CH_2} \xrightarrow[(ii)NaSO_3H]{(i)OsO_4} \underset{(A)}{CH_2OH-CHOH-CH_2OH} \xrightarrow{KHSO_4} \underset{(B)}{CH_2=CH-CHO}$$
$$\xrightarrow{Zn-Hg/\text{ conc. HCl}} \underset{(C)}{CH_2=CH-CH_3} \xrightarrow[CCl_4]{NBS/hv} \underset{(D)}{CH_2=CH-CH_2Br} \xrightarrow{(CH_3)_2CuLi} \underset{(E)}{CH_2=CH-CH_2-CH_3}$$

Q.30 Answer:

$2.8-\log(0.54)$

Solution:

$Ca(LaC)_2 \rightarrow Ca^{2+}+2LaC^-$

Concentration of salt $=\frac{0.6}{2}=0.3M$

Concentration of $\left(LaC^-\right)$ **after dissociation** $=2\times0.3\times0.9=0.54M$pH **after salt hydrolysis may be calculated as,**

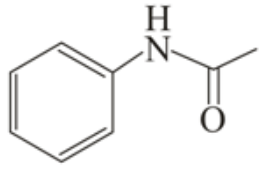
$pH=\frac{1}{2}\left[\left(pK_w+pK_a+\log\left(LaC^-\right)\right]\right.$

$14-5.6=\frac{1}{2}\left[14+pK_a+\log 0.54\right]$

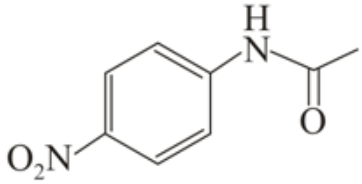
$pK_a=2.8-\log 0.54$

Q.31 Answer:

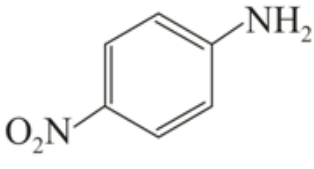
X =



Y =



Z =



Solution:

In step I- The formation of acetanilide takes place in the presence of acetic anhydride.

In step II- The nitration of acetanilide takes place in the presence of nitrating mixture at para position mainly due to the bulkiness of $-NHCOCH_3$ **group.**

In step III- The hydrolysis of $-NHCOCH_3$ **group takes place to regain** $-NH_2$ **group.**

Q.32 Answer:

9.57 L

Solution:

$2NaCl+MnO_2+3H_2SO_4 \rightarrow 2NaHSO_4+MnSO_4+Cl_2+2H_2O$

117 g of $NaCl \equiv 22.4$ L of Cl_2

$50\ \text{g of NaCl} \equiv \frac{22.4}{117} \times 50 = 9.57\ \text{L of } Cl_2 \text{ at STP.}$

Q.33 Answer:

$2n - 2$

Solution:

During electrolysis, two molecules of the potassium salt of monocarboxylic acid combine to form alkane along with the loss of two COO^- group. Hence alkane formed will contain $= 2n - 2$ (no of C−atom).

Q.34 Answer:

A-p,s ; B-q,s,t; C-p; D-r

Solution:

$Fe(CN)_2 + 4KCN \rightarrow K_4[Fe(II)(CN)_6]^{4-}$; **coordination number 6.**

$CuSO_4 + 4NH_3 \rightarrow [Cu(II)(NH_3)_4]^{2+}SO_4^{2-}$; **coordination number 4.**

$3NaF + AlF_3 \rightarrow Na_3[Al(III)F_6]^{3-}$; **coordination number 6.**

$AgCN + KCN \rightarrow K[Ag(1)(CN)_2]^{1-}$; **coordination number 2.**

Q.35 Answer:

$51.08\ \text{kJ mol}^{-1}$

Solution:

Given:

$H_2O(s) \rightleftharpoons H_2O(l)\ ; \quad \Delta H = 6.01\ \text{kJ/mol} \ldots\ldots \text{(i)} H_2O(l) \rightleftharpoons H_2O(g)\ ; \quad \Delta H = 45.07\ \text{kJ/mol} \ldots\ldots \text{(ii)}$

So,

$H_2O(s) \rightleftharpoons H_2O(g)\ \ \Delta H_{sub.} = \Delta H_{fus.} + \Delta H_{vap.} \qquad = 6.01 + 45.07 \qquad = 51.08\ \text{kJ/mol}$

Q.36 Answer:

It is a non-reducing sugar

Solution:

It is a reducing sugar. The free aldehyde group can be produced at $C - 1$ of second glucose in solution and it shows reducing properties.

Q.37 Answer:

Colour of chromate is yellow and dichromate is red.

Solution:

Chromate is yellow and dichromate is orange.

Q.38 Answer:

Hg_2Cl_2 and $HgCl_2$

Solution:

$$Hg_2Cl_2 + 2NH_4OH \longrightarrow \underset{\text{(Black)}}{\left[H_2N - Hg - \underset{(B)}{Cl} + Hg\right]} \downarrow + NH_4Cl + 2H_2O\ HgCl_2 + 2NH_4OH \longrightarrow \underset{\text{(White)}}{Hg(NH_2)Cl} + NH_4Cl + H_2O\ HgCl_2 + 2KI \longrightarrow \underset{\text{(Red)}}{HgI_2} \downarrow + 2KCl$$

$HgCl_2 + 2KI \longrightarrow \underset{\text{(Red)}}{HgI_2} \downarrow + 2KCl$

$HgI_2 + 2KI \text{ (excess)} \longrightarrow K_2(HgI_4)$

Q.39 Answer:

m-nitrophenol > o-nitrophenol

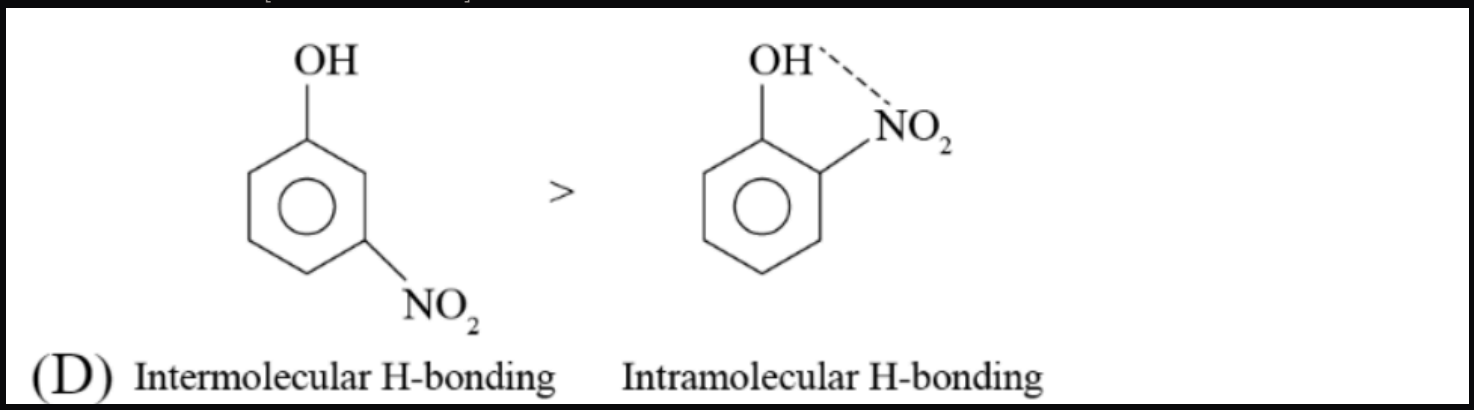
Solution:

Here option D is correct while rest are incorrect as follows

(A) $T_2 > D_2 > H_2$ [BP $\propto$ **mol. wt.**]

(B) n-pentane > neo-pentane [BP $\propto$ VWf $\propto$ **contact area** $\propto \frac{1}{\text{Branching}}$]

(C) $Xe > Ar > He$ [BP $\propto$ **mol. wt.**]



Q.40 Answer:

4, 3, 2, 1

Solution:

vant Hoff factors for urea, $NaCl$, Na_2SO_4 **and** Na_3PO_4 **are** 1, 2, 3 **and 4 respectively. As** $\Delta T_f \propto im$, **so** ΔT_f **for urea,** $NaCl$, Na_2SO_4 **and** Na_3PO_4 **are proportional to** 1, 2, 3 **and 4 respectively. Hence, freezing point order is** $Na_3PO_4 < Na_2SO_4 < NaCl <$ **Urea**

Q.41 Answer:

$\frac{9a_0}{2Z}$

Solution:

Radial nodes occurs where probability of finding electron is zero.

$\psi^2 = 0$ or $\psi = 0$

$\sigma^2 - 5\sigma + 6 = 0$

$\sigma^2 - 3\sigma - 2\sigma + 6 = 0$

$\sigma(\sigma - 3) - 2(\sigma - 3) = 0$

$\sigma = 2$ or 3

For maximum distance, $\sigma = 3$

$3 = \frac{2}{3}\frac{rZ}{a_0}r$

$r = \frac{9a_0}{2Z}$

Q.42 Answer:

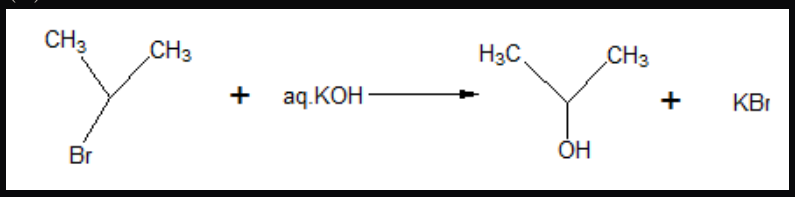
(a) is elimination, (b) is substitution and (c) is addition reaction

Solution:

(a) $CH_3CH_2CH_2Br$ + **alcoholic** $KOH \rightarrow CH_3CH = CH_2 + KBr + H_2O$

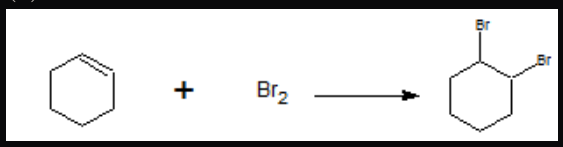
This reaction is elimination reaction.

(b)



This reaction is substitution reaction.

(c)



This reaction is addition reaction.

Q.43 Answer:

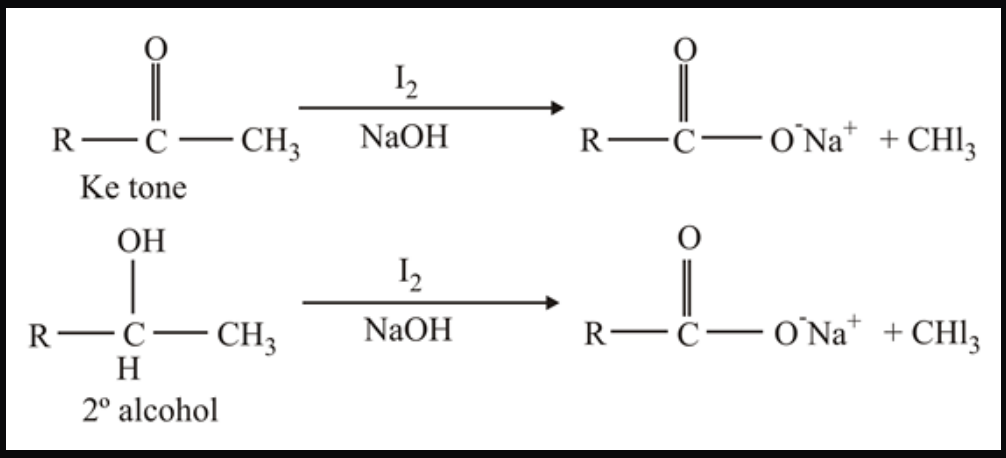
1-Propanol

Solution:

The iodoform test is a test for the presence of carbonyl compounds with the structure $\mathrm{RCOCH_3}$ and alcohols with the structure $\mathrm{RCH(OH)\,CH_3}$.

In an Iodoform test, a solution of $\mathrm{I_2}$ is added to a small amount of the unknown sample, followed by just enough NaOH to remove the color. The formation of a pale yellow precipitate of Iodoform is a positive result.

The reaction is as follows:



Thus, out of the given reagents, $1-$propanol can not show this reaction.

Q.44 Answer:

2.5

Solution:

$\mathrm{A(g) \longrightarrow 2\,B(\,g); k_1 = 6.93 \times 10^{-3}\ min^{-1}}$

$\mathrm{A(\,g) \longrightarrow C(g); k_2 = 6.93 \times 10^{-3}\ min^{-1}}$

$\mathrm{k = (k_1 + k_2)}$ overall velocity constant $= 6.93 \times 2 \times 10^{-3}\ \mathrm{min^{-1}}$

$$t_{1/2} = \frac{0.693 \times 1000}{6.93 \times 2} = 50 \text{ min}$$

i.e., after $= 50$ min

$P_A = 1$ atm

Since $k_1 = k_2$

$P_B = 0.5 \times 2 = 1$ atm **(due to stoichio metric coefficient)**

$P_C = 0.5 \times 1 = 0.5$ atm

Total pressure $= 2.5$ atm.

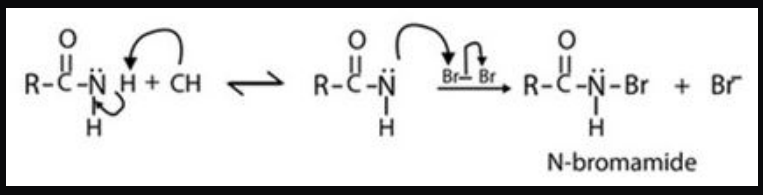
Q.45 Answer:

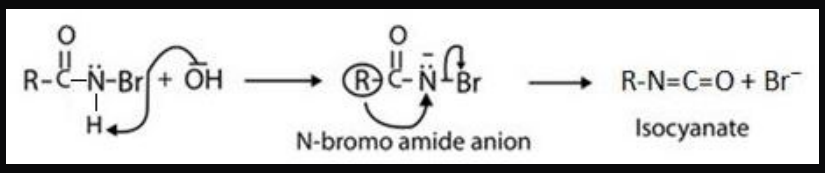
$\mathrm{RCONBr_2}$

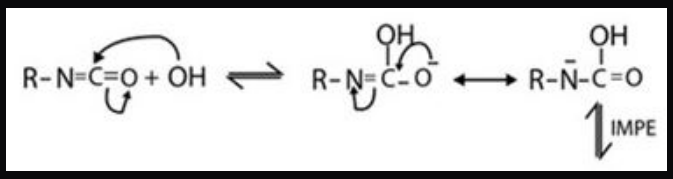
Solution:

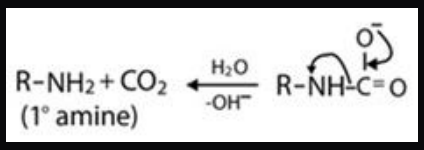
This is Hofmann bromamide reaction and $\mathrm{RCONBr_2}$ does not form in this reaction. See mechanism for it:

$\mathrm{RCO-NH_2 + Br_2 + 4KOH \rightarrow R-NH_2 + K_2CO_3 + 2KBr + 2H_2O}$





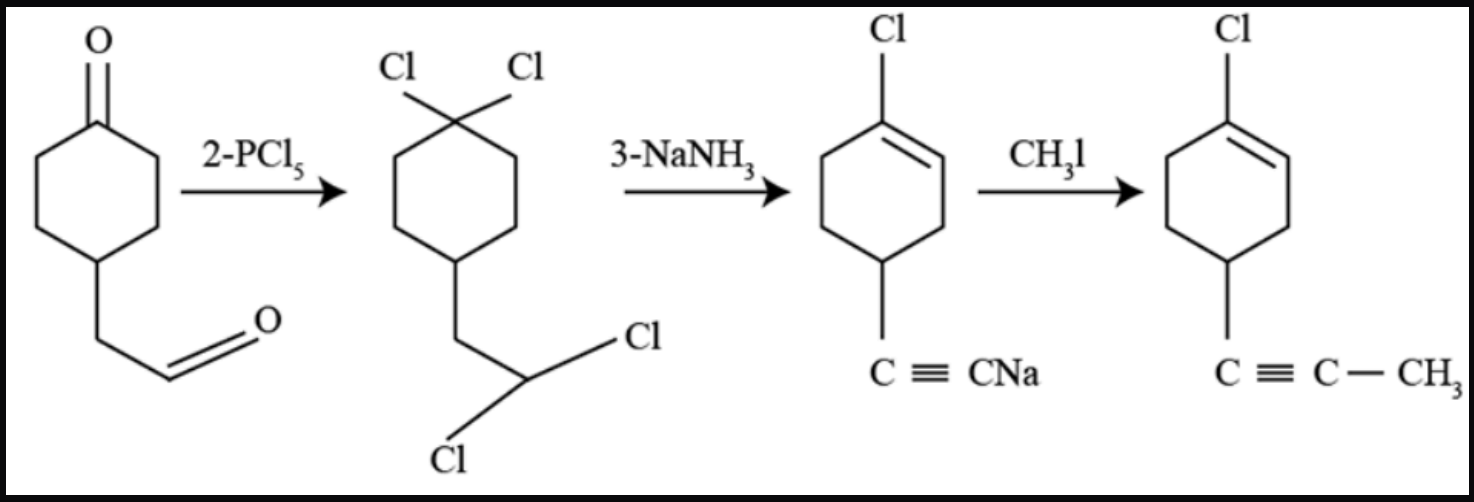




Q.46 Answer:

—

Solution:



Here, $X = 2$ and $Y = 3$

So sum of $X + Y = 2 + 3 = 5$.

Q.47 Answer:

—

Solution:

For adiabatic condition.

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

or $\frac{T_2}{T_1} = \left(\frac{1}{27}\right)^{1.33-1} = \left(\frac{1}{27}\right)^{0.33}$

$= \left(\frac{1}{27}\right)^{1/3} = \frac{1}{3}$

$T_2 = \frac{1}{3} \times T_1$

or $T_2 = \frac{1}{3} \times 300$

$= 100\ \text{K}$

From first law of thermodynamics, $\Delta E = q + W$

$W = -\Delta E$

$(\because q = 0 \text{ for adiabatic expansion }) = -C_V\,(T_2 - T_1)$

$= -6(100 - 300)$

$= +1200\text{ca}$

Q.48 Answer:

—

Solution:

We know, NaCl **will dissociate as:**

$\text{NaCl} \xrightarrow{100\%} \text{Na}^+ + \text{Cl}^-$

Moles of $\text{Na}^+ = 1 \times 0.05 = 0.05\ \text{mol}$

Moles of $\text{Cl}^- = 1 \times 0.05 = 0.05\ \text{mol}$

So, total number of moles of ions $= 2 \times 0.05 = 0.1\ \text{mol.}$

Now,

Mass of non-electrolyte$= \text{x g}$

Molar mass of non-electrolyte $= 200\ \text{g}$

Therefore, moles of non-electrolyte $= \frac{\text{x}}{200}\ \text{mol}$

Total moles of solute into solution $= \left(\frac{\text{x}}{200} + 0.1\right)\ \text{mol}$

As volume of solution is $1\ \text{lit}$

So, Concentration(C) **of solution**$= \left(\frac{\text{x}}{200} + 0.1\right)\ \text{M.}$

Applying the concept of Osmotic Pressure i. e.

$\pi = \text{CRT}$

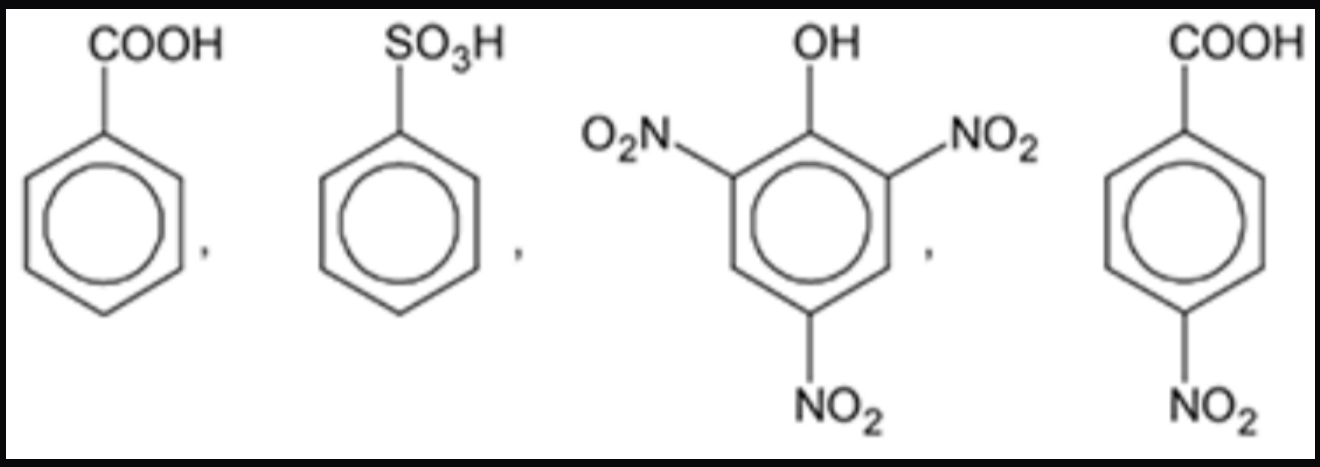
$4.92 = \left(\frac{\text{x}}{200} + 0.1\right) \times 0.0821 \times 300$

$\Rightarrow \frac{4.92}{300 \times 0.0821} = \frac{\text{x}}{200} + 0.1 \Rightarrow 0.2 = \frac{\text{x}}{200} + 0.1 \text{ or, } \text{x} = 20\ \text{g.}$

Q.49 Answer:

—

Solution:



Q.50 Answer:

—

Solution:

MnO is basic in nature. MnO_2, Mn_2O_3 and Mn_3O_4 are amphoteric in nature. These compounds show acidic as well as basic nature.MnO_3 and Mn_2O_7 are acidic in nature.

Q.51 Answer:

reflexive and symmetric both

Solution:

$\because (x,\ y) \in R \Rightarrow x^y = y^x$

$\therefore (x,\ x) \in R$ **as** $x^x = x^x\ \forall\ x \in I - \{0\}$

$\therefore R$ **is reflexive**

Now $(x,\ y) \in R \Rightarrow x^y = y^x \Rightarrow y^x = x^y$

$\Rightarrow (y,\ x) \in R\ \ \therefore R$ **is symmetric**

Now $(x,\ y) \in R \Rightarrow x^y = y^x$ **and** $(y,\ z) \in R \Rightarrow y^z = z^y$

$\because x^y = y^x \Rightarrow y = x^{\frac{y}{x}}$

$\Rightarrow y^z = z^x \Rightarrow \left(x^{\frac{y}{x}}\right)^z = z^x \Rightarrow x^{\frac{yz}{x}} = z^x$

$\Rightarrow x^{yz} = z^{x^2} \Rightarrow \left(x,\ z\right) \notin R$

R **is not transitive.**

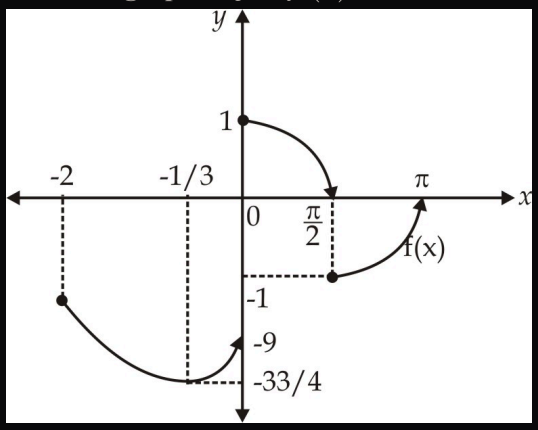
Q.52 Answer:

has a local minima at $x = 0$ and a local maxima at $x = \frac{\pi}{2}$

Solution:

$$f'(x) = \begin{cases} 3x^2 + 2x - 10 & , \quad -2 \le x < 0 \\ \cos x & , \quad 0 < x < \frac{\pi}{2} \\ -\sin x & , \quad \frac{\pi}{2} \le x < \pi \end{cases}$$

Draw the graph of $y = f'(x)$



The graph of $f'(x)$ **changes its sign negative to positive about** $x = 0$ **and positive to negative about** $x = \frac{\pi}{2}$**.**

$\therefore x = 0$ **is the point of local minima and** $x = \frac{\pi}{2}$ **is the point of local maxima.**

Q.53 Answer:

± 1

Solution:

Given,

line $y = x\tan\theta$ **cuts the curve** $x^3 + xy^2 + 2x^2 + 2y^2 + 3x + 1 = 0$ **... (i)**

at three points A, B**and** C**.**

Let us assume $A\left(x_1, x_1\tan\theta\right), B\left(x_2, x_2\tan\theta\right)$ **and** $C\left(x_3, x_3\tan\theta\right)$

$\Rightarrow OA = \sqrt{x_1^2 + x_1^2 \tan^2\theta}$

$= x_1\sqrt{1 + \tan^2\theta} = x_1 \sec\theta$

$= x_1\sqrt{1 + \tan^2\theta} = x_1 \sec\theta$

Similarly $OB = x_2 \sec\theta$ **and** $OC = x_3 \sec\theta$

$\because OA, OB, OC$ **in** H.P

$\Rightarrow OB = \dfrac{2 \cdot (OA)(OC)}{OA + OC}$

$\Rightarrow x_2 \cdot \sec\theta = \dfrac{2x_1 \cdot x_3 \sec^2\theta}{(x_1 + x_3)\sec\theta}$

$\Rightarrow x_1x_2 + x_2x_3 = 2x_1x_3$

$\Rightarrow x_1x_2 + x_2x_3 + x_3x_1 = 3x_1x_3 \quad \ldots(ii)$

$\Rightarrow x_2 \cdot \sec\theta = \frac{2x_1 \cdot x_3 \sec^2\theta}{(x_1 + x_3)\sec\theta}$

$\Rightarrow x_1x_2 + x_2x_3 = 2x_1x_3$

$\Rightarrow x_1x_2 + x_2x_3 + x_3x_1 = 3x_1x_3$ **... (ii)**

Now, substitute $y = x\tan\theta$ **in equation** (i)**, we get**

$x^3 + x^3\tan^2\theta + 2x^2 + 2x^2\tan^2\theta + 3x + 1 = 0$

$\Rightarrow x^3\sec^2\theta + 2x^2\sec^2\theta + 3x + 1 = 0 \quad \ldots(iii)$

$\Rightarrow x^3\sec^2\theta + 2x^2\sec^2\theta + 3x + 1 = 0$ **...(iii)**

Now, the roots of the equation (iii) **will be** x_1, x_2, x_3**.**

$\Rightarrow x_1 \cdot x_2 + x_2 \cdot x_3 + x_3 \cdot x_1 = \frac{3}{\sec^2\theta}$ **...(iv)**

$\Rightarrow x_1 \cdot x_2 \cdot x_3 = -\frac{1}{\sec^2\theta}$ **...(v)**

From equations (ii) **and** (iv), $3x_1x_3 = \frac{3}{\sec^2\theta}$

$\Rightarrow x_1x_3 = \frac{1}{\sec^2\theta}$ **...(vi)**

From equations (v) **and** (vi), $x_2 = -1$

$\because x_2$ **is one of the roots of the equation (iii), so it will satisfy the equation.**

$\Rightarrow -\sec^2\theta + 2\sec^2\theta - 3 + 1 = 0$

$\Rightarrow \sec^2\theta = 2$

$\Rightarrow \sec\theta = \pm\sqrt{2}$

$\Rightarrow \tan\theta = \pm 1.$

Q.54 Answer:

$\frac{6}{55}$

Solution:

$n(S) = {}^{11}C_2 = 55$

Set of Event $E = \{(0,3),(0,6),(0,9),(3,6)(3,9),(6,9)\}$

So, $n(E) = 6 \quad \Rightarrow P(E) = \frac{6}{55}$

Q.55 Answer:

21

Solution:

$\because (f(x) - 3)\left(f\left(\frac{1}{x}\right) - 3\right) = 4$

$\therefore \quad f(x) - 3 = \pm 2x^n \Rightarrow f(x) = 3 \pm 2x^n$

$\because \quad f(2) = 3 + 2 \cdot 2^n \doteq 11 \Rightarrow 2^n = 4 \Rightarrow n = 2$

$\therefore \quad f(x) = 3 + 2x^2 \Rightarrow f(3) = 21$

Q.56 Answer:

$\sum_{i=1}^{k} \Delta_i = 0$

Solution:

The number of third-order determinants = the number of arrangements of nine different numbers in nine places = $9!$
Corresponding to each determinant made, there is a determinant obtained by interchanging two consecutive rows (or columns). So, the sum of this pair will be 0.
$\therefore$ the sum of all determinants
$= 0 + 0 + 0 + ...= 0$

Q.57 Answer:

$\mathrm{I}_1 = \mathrm{I}_2$

Solution:

In I_2 substitute $x = \frac{1}{t} \Rightarrow dx = -\frac{dt}{t^2}$
So, $\mathrm{I}_2 = -\int_1^0 \frac{\ln t}{t^2+4t+1} \cdot t^2\left(\frac{-dt}{t^2}\right) = \int_0^1 \frac{(-\ln t)}{t^2+4t+1} dt = \mathrm{I}_1$ $\{\text{as } |\ln t| = -\ln t \text{ in } (0,1)\}$

Q.58 Answer:

$25,\ 20$

Solution:

In probability theory and statistics, the coefficient of variation (CV), also known as relative standard deviation (RSD), is a standardised measure of dispersion of a probability distribution or frequency distribution. It is often expressed as a percentage, and is defined as the ratio of the standard deviation to the mean.
Coefficient of variation$= \frac{\sigma}{\overline{x}} \times 100$.
Clearly, $70 = \frac{17.5}{\overline{x}_1} \times 100$.
$\Rightarrow \overline{x}_1 = 25$ and $90 = \frac{18}{\overline{x}_2} \times 100$.
$\therefore\ \overline{x}_2 = 20$

Q.59 Answer:

$(2,\ 4,\ 6)$

Solution:

Point of intersection of lines $x = y = z$ and $x = \frac{y}{2} = \frac{z}{3}$ is $(0,\ 0,\ 0)$
Angle between lines is $\cos\theta = \frac{(1)(1)+(1)(2)+(1)(3)}{\sqrt{1+1+1}.\sqrt{1+2^2+3^2}} = \frac{6}{\sqrt{42}}$
$\Rightarrow \sin\theta = \frac{\sqrt{6}}{\sqrt{42}}$
Let a point on line $x = \frac{y}{2} = \frac{z}{3}$ is $A(\lambda,\ 2\lambda,\ 3\lambda)$
$x = y = z$ passes through $B(1, 1, 1)$ and third line also passes through $B(1,\ 1,\ 1)$
$\therefore$ Area of $\Delta OAB = \frac{1}{2} OA.\, OB \sin\theta = \frac{1}{2}\sqrt{3}\,\lambda\sqrt{14}\ \ \frac{\sqrt{6}}{\sqrt{42}} = \sqrt{6}$
$\therefore\ \lambda = 2$
$\therefore$ Required coordinate $= (2,\ 4,\ 6)$.

Q.60 Answer:

no x

Solution:

$\mathrm{f(x)} = \{\mathrm{x}\} \sin(\pi[\mathrm{x}])$
$= \{\mathrm{x}\} \sin$ **(integral multiple of π)**

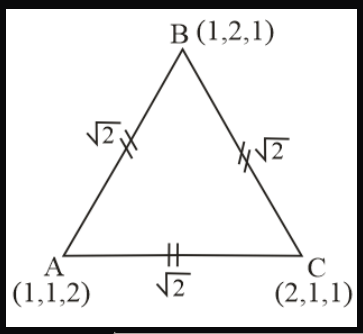
$= 0$

Hence, $f(x)$ is continuous for all x.

Q.61 Answer:

$\frac{3}{\sqrt{6}}$

Solution:



$AB = \sqrt{(1-1)^2 + (2-1)^2 + (1-2)^2}$

$AB = \sqrt{2}$ units

Similarly, $BC = \sqrt{2}$ units **and** $CA = \sqrt{2}$ units.

Hence, ΔABC **is an equilateral triangle.**

Hence, the distance of the orthocentre O **from the sides is equal to the distance of the incentre from the sides, that is, the inradius of the triangle.**

Since, in an equilateral triangle, the orthocentre and incentre coincides.

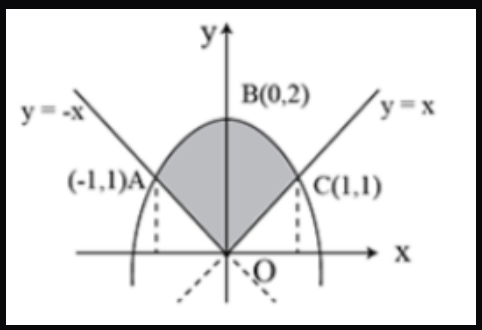
$\therefore l_1 = l_2 = l_3 =$ **inradius** $\Rightarrow r = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4}\left(\sqrt{2}\right)^2}{\frac{3}{2}\left(\sqrt{2}\right)} = \frac{1}{\sqrt{6}}$

$\Rightarrow (l_1 + l_2 + l_3) = \frac{3}{\sqrt{6}}$

Q.62 Answer:

$\frac{7}{3}$

Solution:



We first draw the given curves

The first curve $x^2 - y^2 = 0 \Rightarrow y = \pm x$

represents a pair of straight lines with slopes 1 **and** -1 **passing through origin. The second curve**

$\Rightarrow x^2 + y - 2 = 0 \Rightarrow x^2 = -y + 2$

$\Rightarrow x^2 = -(y - 2)$

represents a parabola with vertex $(0, 2)$ **axis as** $y-$**axis and concavity downwards. Both the curves are plotted in the figure and the required area is shown by the shaded region. The points** A **and** C **are the points of intersection of** $y^2 = x^2$ **with** $x^2 + y - 2 = 0$.

Solving the two equations, we get $y^2 + y - 2 = 0$ **[putting value of** $x^2 = y^2$ **]**

$\Rightarrow (y + 2)(y - 1) = 0$

giving $y = -2$ **and** 1**, but** $\mathbf{y = -2}$ **is discarded as the required area is above the** x**-axis.**

$\therefore y=1 \Rightarrow x=\pm 1$

The points A and C are respectively $(-1,1)$ and $(1,1)$ now due to symmetry Area of the bounded region OABCO

$=2\times$ **Area** $\mathrm{OBCO}=2\times\int_0^1\left[\left(2-\mathrm{x}^2\right)-\mathrm{x}\right]\mathrm{dx}$

[Since $y=2-x^2$ is the upper curve and $y=x$ is the lower curve]

$=2\left[2x-\frac{x^3}{3}-\frac{x^2}{2}\right]_0=2\left[2-\frac{1}{3}-\frac{1}{2}\right]=\frac{7}{3}$

Q.63 Answer:

I

Solution:

Given, P is orthogonal

$\therefore \quad p^TP=I$

and $\quad Q=PAP^T$

Now, $\quad B=P^TQ^{1000}P=P^T\left(PAP^T\right)^{1000}P$ **[from Eq. (ii)]**

$=P^TPAP^T\left(PAP^T\right)^{999}P$

$=IAP^T\cdot PAP^T\left(PAP^T\right)^{998}P$

$=AIAP^T\left(PAP^T\right)^{998}P$

$=A^2P^TPAP^T\left(PAP^T\right)^{997}P$

$=A^2\left[AP^T\left(PAP^T\right)^{997}P\right.$

$=A^3P^T\left(PAP^T\right)^{997}P$

$\cdots \qquad \cdots$

$=A^{1000}P^TP=A^{1000}=I \quad [\because A$ **is involutory** $]$

Hence, $\quad B^{-1}=I^{-1}=I$

Q.64 Answer:

616

Solution:

EHINTZ

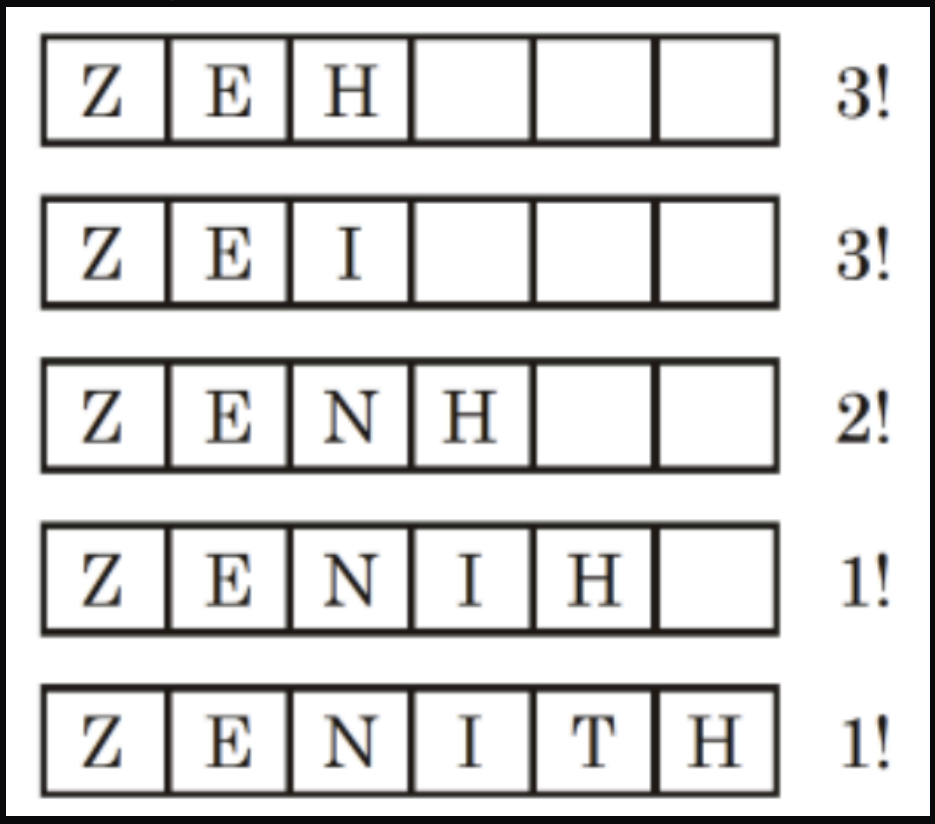
words starting with E is 5 !

words starting with H is 5 !

words starting with I is 5 !

words starting with N is 5 !

words starting with T is 5 !



The position of word ZENITH is 616

Q.65 Answer:

$\log_e |xy| = \sin^{-1} x + C$

Solution:

$$xdy + ydx = \frac{xydx}{\sqrt{1-x^2}}$$

$$\frac{d(xy)}{xy} = \frac{dx}{\sqrt{1-x^2}}$$

Integrate both sides we get $\log_e xy = \sin^{-1} x + C$

Q.66 Answer:

$-\hat{i} - 8\hat{j} + 2\widehat{k}$

Solution:

$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$

$\vec{r} \times \vec{b} - \vec{c} \times \vec{b} = \vec{0}$

$\left(\vec{r} - \vec{c}\right) \times \vec{b} = \vec{0}$

$\left(\vec{r} - \vec{c}\right) \ // \ \vec{b}$

$\vec{r} - \vec{c} = t\vec{b}$

$\vec{r} = \vec{c} + t\vec{b}$

$\vec{r}-\vec{a}=\vec{c}.\vec{a}+t\left(\vec{b}.\vec{a}\right)$

$\frac{-\left(\vec{c}.\vec{a}\right)}{\vec{b}.\vec{a}}=t$

$\vec{r}=\vec{c}=\left(\frac{\vec{c}.\vec{a}}{\vec{b}.\vec{a}}\right)$

Q.67 Answer:

$p^2=q$

Solution:

Given polynimial is, $x^3+3px^2+3qx+r=0\ \ldots.(\mathrm{i})$

z_1,z_2,z_3 **are roots of** (i)

$\Rightarrow\ z_1+z_2+z_3=-3p,\ \&\ \ z_1\cdot z_2+z_2.z_3+z_3\cdot z_1=r$

For equilateral Δ **we know that**

$z_1^2+z_2^2+z_3^2=z_1z_2+z_2z_3+z_3z_1$

$\Rightarrow\ \left(z_1+z_2+z_3\right)^2=3(z_1z_2+z_2z_3+z_3z_1)$

$\Rightarrow(-3p)^2=3\times3q$

$p^2=q$

Q.68 Answer:

$(40)^{20}$

Solution:

$13,\ a_1,\ a_2,\ a_3,\ \ldots.a_{20},\ 67$ **are in AP.**

$\therefore\ \ a_1+a_2+a_3+...+a_{20}=20\left(\frac{13+67}{2}\right)=800$

Now, $\mathrm{AM}\geq\mathrm{GM}$

$\because\ \ \frac{a_1+a_2+a_3+...a_{20}}{20}\geq\left(a_1a_2a_3...a_{20}\right)^{\frac{1}{20}}$

$\Rightarrow\left(\frac{800}{20}\right)\geq\left(a_1a_2a_3...a_{20}\right)^{\frac{1}{20}}$

or, $\ \ a_1a_2a_3...a_{20}\leq(40)^{20}$

$\therefore$**The maximum value of** $a_1\cdot a_2\cdot a_3\cdot\ldots.a_{20}$ **is** $(40)^{20}$**.**

Q.69 Answer:

$-\frac{2}{11}$

Solution:

Ellipse: $\frac{x^2}{45}+\frac{y^2}{20}=1$

Foci: $(\pm5,\ 0)$

Let's say $\theta=\cos^{-1}\left(\frac{-1}{\sqrt{5}}\right)\Rightarrow\cos\theta=\frac{-1}{\sqrt{5}}\Rightarrow\tan\theta=-2$

Now, equation of line passing through $(-5,\ 0)$ **and of slope** -2 **can be written as** $y=-2(x+5)$**or** $y=-2x-10$

Now, if we solve this line with given ellipse equation

$\frac{x^2}{45}+\frac{(-2x-10)^2}{20}=1$

This gives us point of intersection above x**-axis at** $(-6,\ 2)$**. (Point** $(-3,4)$ **will be rejected as slope of line joining it with focus will be 2 but slope of line given is** -2**).**

Now, since line meets the ellipse above x**-axis at** $(-6,\ 2)$**. Therefore, it is also the point from where ray got reflected.**

From reflection property of ellipse, we can say that reflected ray passes through the other focus whose coordinates are $(5,\ 0)$**.**

Hence, slope $= \frac{2-0}{-6-5} = -\frac{2}{11}$

Q.70 Answer:

${}^{2n-1}C_{n-1}$

Solution:

There is no loss of generality in considering the order of bi's as $b_1 < b_2 < b_3 \cdots < b_n$ **also given that** $f(a_1) \leq f(a_2) \cdot f(a_{2n})$. **Now suppose number of preimages of every** b_i **are** x_i **in numbers. Therefore**

$x_1 + x_2 + \cdots + x_n = 2n$ **where** $1 \leq x_i \leq n+1 \rightarrow (1)$

Number of solutions of (1) **is** ${}^{2n-1}C_{n-1}$ **or** ${}^{2n-1}C_n$

Q.71 Answer:

—

Solution:

Let AB **and** CD **be two rods of length** a **and** b **sliding along** $x-$ **axis and** $y-$**axis respectively.**

Let the equation of circle through A, B, C **and** D.

$x^2 + y^2 + 2gx + 2fy + c = 0$

$AB = x-$ **intercept and circle**

$a = 2\sqrt{g^2 - c}$ **i.e.** $a^2 = 4\left(g^2 - c\right) \quad \ldots (1)$

$CD = y-$ **intercept and circle**

$b = 2\sqrt{f^2 - c}$ **i.e.** $b^2 = 4\left(f^2 - c\right) \quad \ldots (2)$

From (1) **and** (2)

$\left(a^2 - b^2\right) = 4\left(g^2 - f^2\right)$

Locus of centre is $4\left(x^2 - y^2\right) = a^2 - b^2$

$k = 4$

Q.72 Answer:

—

Solution:

Since, $\alpha^2 - 2\alpha + 3 = 0$

$\Rightarrow \alpha^3 - 3\alpha^2 + 5\alpha - 2 = \alpha\left(\alpha^2 - 2\alpha + 3\right) - \left(\alpha^2 - 2\alpha + 3\right) + 1 = 1$

And $\beta^2 - 2\beta + 3 = 0$

$\Rightarrow \beta^3 - \beta^2 + \beta + 5 = \beta\left(\beta^2 - 2\beta + 3\right) + \left(\beta^2 - 2\beta + 3\right) + 2 = 2$

So, the required equation is $x^2 - (2+1)x + 2.1 = 0$

$\Rightarrow x^2 - 3x + 2 = 0$

Sum of roots $= 3$

Q.73 Answer:

—

Solution:

If n **is odd, then** $3^n = 4\lambda_1 - 1\ ,\ 5^n = 4\lambda_2 + 1\ ,\ 2^n + 3^n + 5^n$ **is divisible by four i.e.** n **can take** 49 **different values excluding** 1.

If n **is even, then** $3^n = 4\lambda_1 + 1\ ,\ 5^n = 4\lambda_2 + 1\ ,\ 2^n + 3^n + 5^n$ **is not divisible by four as** $2^n + 3^n + 5^n$ **will be in the form of** $4\lambda + 2$. **Thus total number of ways of selecting** $n = 49$.

$\Rightarrow \frac{49}{7} = 7$

Q.74 Answer:

—

Solution:

$A^T = A, B^T = -B$

$(A+B)(A-B) = (A-B)(A+B)$

$\Rightarrow A^2 - AB + BA - B^2 = A^2 + AB - BA - B^2$

$\Rightarrow AB = BA \ldots\ldots$ (i)

Now, $(AB)^T = B^T A^T = -BA = -AB = (-1)^k AB$

$\Rightarrow k$ **is an odd number**

$\Rightarrow k = 3, 5, 7, 9$

Sum $= 24$

Q.75 Answer:

—

Solution:

Graph of $y = 4\cos^{-1}(-|x|)$

From the graph it is Clear that $\mathrm{k} \in (2\pi, 4\pi]$

$\therefore$ **integral values of** k **are** $7, 8, 9, 10, 11, 12$

